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Self-testing for pregnancy: a systematic review

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Introduction: While urine pregnancy tests are available over-the-counter in high- and middle-income settings, in many low-income settings they are often financially inaccessible outside of public health services. More widespread efforts to provide home pregnancy testing could support greater self care for sexual and reproductive health.

Methods: We conducted a systematic review of studies examining pregnancy self-testing effectiveness, values and preferences of end users and health-care providers, and cost. Standardized methods were used to search, screen, and code included studies. Meta-analysis was conducted using random effects models and findings were summarized in GRADE evidence profile tables.

Results: For pregnancy self-test effectiveness, four randomized trials following individuals after medical abortion showed no difference in loss to follow-up with home pregnancy self-testing compared to return clinic visits (meta-analysis risk ratio: 0.479, 95% confidence interval: 0.155 to 1.480). One additional trial of community health workers offering home pregnancy tests to clients showed a significant increase in pregnancy knowledge and antenatal counseling. There was no data on missed ongoing pregnancy; self-efficacy, self-determination, autonomy, and empowerment; mental health and well-being; adverse events and social harms; or device-related issues. Eighteen diverse values and preferences studies found that most individuals supported pregnancy self-testing because of their quick results, convenience, confidentiality/privacy, cost, and accuracy. Most individuals receiving pregnancy tests for post-abortion home management said they preferred this option. No studies reported cost data.

Conclusion: Pregnancy self-testing is acceptable and valued by end users. Expanded use in the health system, including in post-abortion care and community health worker programs, may lead to improved sexual and reproductive health outcomes. In the current context of the COVID-19 pandemic, this should help maintain the continuity of essential sexual and reproductive health services.

Keywords: Pregnancy test, self care, systematic review, values and preferences, post-abortion care, community health workers, sexual and reproductive health

Systematic review registration number: PROSPERO CRD42021231656

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Strengths and limitations of this study

- Urine pregnancy tests have been shown to have high sensitivity and specificity. While pregnancy tests are available over-the-counter in high and middle-income settings, in many low-income settings they are often financially inaccessible outside of public health services.
- This systematic review identified six effectiveness studies, 18 values and preferences studies, and zero cost studies about pregnancy self-testing.
- For effectiveness, four randomized trials following individuals after medical abortion showed no difference in loss to follow-up with home pregnancy self-testing compared to return clinic visits (meta-analysis risk ratio: 0.479, 95% confidence interval: 0.155 to 1.480). One additional trial of community health workers (CHWs) offering home pregnancy tests to clients showed a significant increase in pregnancy knowledge, and antenatal counseling.
- Globally diverse values and preferences studies found that most individuals supported pregnancy self-testing because of their quick results, convenience, confidentiality/privacy, cost, and accuracy. Most individuals receiving home pregnancy tests for post-abortion management preferred this option.
- Pregnancy self-tests are effective and acceptable approaches to improving sexual and reproductive health in diverse health system settings.

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Introduction

Urine tests for pregnancy measure the presence of human chorionic gonadotropin (hCG), and are widely used to detect pregnancy in both home and clinical settings. Urine pregnancy tests have evolved substantially since they were first developed 100 years ago.[1] The current generation of tests have 100% sensitivity and specificity in detecting hCG at concentrations of 25 milli-IU/ml or more, and thus are able to detect pregnancy as early as one day after a missed menstrual period.[1] However, the correct use of urine pregnancy tests varies with user characteristics. A systematic review published in 1998 identified five studies that evaluated the diagnostic efficiency of home pregnancy tests.[2] Sensitivity ranged from 0.52 to 1.0 across tests. A gradient of sensitivity was identified by user group. Sensitivity was highest in studies where urine samples obtained by the investigators were tested by volunteers (sensitivity: 0.91, 95% confidence interval [CI], 0.84-0.96) and lower in actual patients who performed the test on their own urine samples (sensitivity 0.75 [95% CI, 0.64-0.85]). Similarly, the test effectiveness score was 2.75 (95% CI, 2.3-3.2) for studies where subjects were volunteers but 0.82 (95% CI, 0.4-1.2) for studies with actual patients.

Providing pregnancy tests for home use may have a range of benefits for different populations. In Madagascar, randomized trial data has shown that providing pregnancy tests to community health workers for home distribution can increase both engagement in antenatal care services[3] and contraceptive services, since a negative pregnancy test is necessary before initiating some contraceptive methods.[4] Home pregnancy tests have been shown to be an acceptable and feasible option for follow-up among couples undergoing assisted reproduction.[5] There is also evidence supporting the efficacy, safety, and acceptability of urine pregnancy tests to confirm the effectiveness of a medical abortion instead of an ultrasound.[6, 7] While urine pregnancy self-tests are available over-the-counter in many high- and middle-income settings, in many low-income settings, they may be financially inaccessible to most people outside of public health services, or unavailable altogether, leading individuals with the sole option of health facility-based blood tests to confirm pregnancy. For example, most countries in the Eastern Mediterranean Region have pregnancy self-testing widely available in private pharmacies, particularly in urban settings, and mainly used by the upper socio-economic class due cost and knowledge.[8]

Many people in resource-constrained settings are not able to decide if, how many, and when to have children; increased access to self-care interventions, such as pregnancy self-tests, could support their health decision-making. More widespread efforts to provide pregnancy self-tests that can be used at home or another preferred location could support increased autonomy of individuals as well as support multiple programmatic settings to advance sexual and reproductive health and rights.

We sought to review the literature on self-testing for pregnancy as an additional option to facility-based testing. We conducted this systematic review in the context of expanding the evidence base of the 2019 WHO's normative guidance on self-care interventions[9] to include new considerations related to sexual and reproductive health. This review was also conducted in the context of the COVID-19 pandemic which saw disruptions of essential sexual and reproductive health services. The restrictive measures taken to prevent the spread of COVID-19,

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particularly lockdowns, have resulted in an increased need for over-the-counter availability of self-care interventions.

Methods

This review addressed the question: should self-testing for pregnancy be available as an additional option to health facility-based testing? We reviewed the extant literature in three areas relevant to answering this question: effectiveness of the intervention, values and preferences of end users and providers, and cost information. The review followed PRISMA guidelines,[10] and the protocol was published on PROSPERO (CRD42021231656). Ethical approval was not required for this systematic review, since all data came from published articles.

Effectiveness review

The effectiveness review was designed according to the population-intervention-comparison-outcomes (PICO) format as follows: Should self-testing for pregnancy be available as an additional option to health facility-based testing?

Population: Individuals seeking pregnancy testing
Intervention: Urine self-testing for pregnancy
Comparison: Healthcare provider-led testing for pregnancy (health facility or community clinic with either urine and/or serum test for pregnancy)
Outcomes:
(1) Missed ongoing pregnancy
(2) Appropriate clinical follow-up (counseling, antenatal care visit(s), contraceptive services, abortion services, etc.)
(3) Gestational age at pregnancy awareness (knowledge of pregnancy) and at presentation for antenatal care or abortion
(4) Self-efficacy, self-determination, autonomy, and empowerment
(5) Mental health and well-being (anxiety, stress, self-harm)
(6) Adverse events and social harms (including stigma, discrimination, coercion, violence [including intimate-partner violence, violence from family members or community members, etc.] and breaches of confidentiality), and whether these harms were corrected/had redress available
(7) Device-related issues (e.g. test failure, problems with manufacturing, packaging, labelling, or instructions for use)

Inclusion criteria

To be included in the review, an article had to meet the following criteria:

1) Study design that compared urine self-testing for pregnancy to healthcare provider-led testing for pregnancy (health facility or community services with either urine and/or serum test for pregnancy). This included both randomized controlled trials, non-randomized controlled trials and comparative observational studies (including prospective controlled cohort studies, cross-sectional studies, controlled before-after

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studies and interrupted time series) that compare individuals who received the intervention to those who did not.
2) Measured one or more of the outcomes listed above.
3) Published in a peer-reviewed journal.

No restrictions were placed based on location of the intervention. No language restrictions were used on the search. Articles in English, French, Spanish, and Chinese were coded directly; articles in other languages were translated.

Search strategy

The following electronic databases were searched through the search date of November 2, 2020: PubMed, CINAHL, LILACS and EMBASE. We searched for ongoing randomized controlled trials (RCTs) through clinicaltrials.gov, the WHO ICTRP, PACTR, and the Australian New Zealand Clinical Trials Registry. Secondary reference searching was also conducted on all studies included in the review. Further, selected experts in the field were contacted to identify additional articles not identified through other search methods.

Search terms

The following search strategy (Pubmed) was adapted for entry into all computer databases. These search terms were used for the main systematic review (PICO question), for the values and preferences review, and for the cost review.

("pregnancy tests" [Mesh] OR "Pregnancy Tests, Immunologic" [Mesh] OR "pregnancy test" [tiab] OR "pregnancy diagnosis" [tiab])

AND

("self care"[Mesh] OR "self-care"[tiab] OR "self-test"[tiab] OR "self-evaluation"[tiab] OR "self-evaluate"[tiab] OR "self-management"[tiab] OR "self-monitor"[tiab] OR "self-manage"[tiab] OR "self-monitored"[tiab] OR "self-managed"[tiab] OR "home"[tiab])

Screening abstracts

Titles, abstracts, citation information, and descriptor terms of citations identified through the search strategy were screened by a member of the senior study staff. Full text articles were obtained of all selected abstracts, and two independent reviewers assessed all full-text articles for eligibility to determine final study selection. Differences were resolved through consensus.

Data extraction and management

Data were extracted independently by two reviewers using standardized data extraction forms. Differences in data extraction were resolved through consensus, and referral to a senior study team member from WHO when necessary.

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The following information was gathered from each included study:

- Study identification: Author(s); type of citation; year of publication
- Study description: Study objectives; location; population characteristics; type of urine pregnancy test; description of self-test access; study design; sample size; follow-up periods and loss to follow-up
- Outcomes: Analytic approach; outcome measures; comparison groups; effect sizes; confidence intervals; significance levels; conclusions; limitations

For randomized trials, risk of bias was assessed using the Cochrane Collaboration's tool for assessing risk of bias.[11] For non-randomized trials but comparative studies, study rigor was assessed using the Evidence Project 8-item checklist for intervention evaluations.[12]

Data analysis

Data were analyzed according to coding categories and outcomes. Where multiple studies reported the same outcome, meta-analysis was conducted using random-effects models to combine risk ratios with the program Comprehensive Meta-Analysis (CMA).

All analyses were stratified by the following categories/subgroups, where possible:

- Pregnancy self-test point of access (e.g. pharmacy, online, community health worker (CHW) distribution)
- Type of pregnancy test
- Population (e.g. age, marital status, rural/urban)
- Vulnerabilities (e.g. obesity, socio-economic status, poverty, disability, literacy/educational level)
- Gestational age
- High-income versus low or middle-income countries

Data were summarized in GRADE Evidence Profile tables using GRADEPro. Where RCT data were available for a given outcome, we prioritized that over observational data.

Values and preferences review

The same search terms were used to search and screen for studies to be included in the values and preferences review. Studies were included in this review if they presented primary data examining preferences of individuals regarding urine self-testing for pregnancy. We focused on studies examining the values and preferences of women and adolescent girls who were self-testing for pregnancy or individuals who were potential candidates for such self-testing, but we also included studies examining the values and preferences of healthcare providers and other stakeholders. In this section, we also considered issues related to ability to access (by age, gender, or other factors, i.e. legal restrictions around who can access e.g. by age, requirement to be married to purchase etc., and practical barriers, e.g., is it harder for adolescents to access self-tests because they are not allowed out on their own, they have no money, etc.), informed decision-making, coercion, confidentiality, self-determination, health decision-making to terminate or maintain pregnancy (and discussion with partner if appropriate), and seeking

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redress. These studies could be qualitative or quantitative in nature, but had to present primary data collection – think pieces and review articles were not included. Values and preferences literature was summarized qualitatively and organized by study design and methodology, location, and population.

Cost review

The same search terms were used to search and screen for studies to be included in the cost review. Studies were included in this review if they presented primary data comparing costing, cost-effectiveness, cost-utility, or cost-benefit of the intervention and comparison listed in the PICOs above, or if they presented cost-effectiveness of the intervention as it relates to the PICO outcomes listed above. Cost literature was summarized qualitatively. Cost literature was classified into four categories (health sector costs, other sector costs, patient/family costs, and productivity impacts) and within each category were organized by study design/methodology, location, and population.

Patient and public involvement

Several of the authors are current or previous users of pregnancy self-tests. Feedback was also received from the WHO patient safety working group. Patients were involved in a global survey of values and preferences conducted to inform the WHO guideline on self-care interventions; they thus play a significant role in the overall recommendation informed by this review.

Results

Our search yielded 414 unique references, of which 62 were retained for full-text review (Figure 1). Ultimately, we identified six that met the inclusion criteria for the effectiveness review, 18 values and preferences studies, and no cost studies.

Figure 1. PRISMA flow chart showing disposition of citations through the search and screening process

Effectiveness review

Overall, six studies met the inclusion criteria for the effectiveness review. This included five RCTs[3, 13-16] and 1 observational study.[17] Table 1 presents descriptive data from the 5 RCTs. Four of the RCTs, conducted in a diverse range of countries (India, Vietnam, Austria, Finland, Norway, Sweden, Moldova, and Uzbekistan) were conducted among individuals receiving medical abortion.[13-16] These RCTs randomized clients to abortion follow-up with home pregnancy testing and a phone call, versus abortion follow-up with the traditional clinic visit, usually with ultrasound confirmation of successful termination. The fifth RCT, conducted in Madagascar, randomized CHWs to receive pregnancy tests to use with their clients versus the standard of care, which the authors said had historically been pregnancy testing available only at clinics.[3]

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Table 1. Description of RCTs included in the effectiveness review

Study	Location	Sample size	Intervention	Comparator	Outcomes
Women who had a medical abortion					
Iyengar et al., 2015	India (Rajasthan: urban and rural)	N=731	Abortion follow-up with home pregnancy testing and phone call	Abortion follow-up with clinic visit	2. Appropriate clinical follow-up
Ngoc et al., 2014	Vietnam (Ho Chi Minh City and Hanoi)	N=1433			
Oppegaard et al., 2015	Austria (Vienna), Finland (Helsinki), Norway (Oslo), Sweden (Stockholm)	N=929			
Platais et al., 2015	Moldova (Chisinau, Balti, and Drochia), Uzbekistan (Tashkent)	N=2400			
Community health workers and their clients					
Comfort et al., 2019	Madagascar (Aloatra Mangoro, Atsinanana and Analanjirofo)	N=506 CHWs and their clients	Providing CHWs with pregnancy tests to give to clients	Pregnancy testing available at clinics	2. Appropriate clinical follow-up 3. Gestational age at pregnancy awareness (knowledge of pregnancy) and at presentation for antenatal care or abortion

The RCTs provided data for two of our PICO outcomes: 2) Appropriate clinical follow-up, and 3) Gestational age at pregnancy awareness (knowledge of pregnancy) and at presentation for antenatal care or abortion. The non-randomized observational study provided data only for the same outcome under appropriate clinical follow-up, so per our protocol, we present only the RCT data here.

Appropriate clinical follow-up was assessed in the four post-abortion RCTs as loss to follow-up, meaning the client did not return for their follow-up visit or was not able to be contacted by phone. In meta-analysis, there was no significant difference between study arms in loss to follow-up (pooled risk ratio: 0.479, 95% confidence interval 0.155 to 1.480) (Figure 2).

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Heterogeneity was substantial, with an I-squared of 87. Stratification by high- vs. low- and middle-income settings did not yield meaningful differences; no further stratifications were available.

Figure 2. Meta-analysis of loss to follow-up (LTFU) among clients following medical abortion, comparing those receiving follow-up at home versus a return clinic visit

Appropriate clinical follow-up was assessed in the CHW RCT as the mean number of clients at risk of pregnancy who received antenatal counselling at their CHW visit per CHW. This was higher in the intervention group (mean difference: 0.39 clients more, 95% CI: 0.14 more to 0.64 more).

Gestational age at pregnancy awareness (knowledge of pregnancy) and at presentation for antenatal care or abortion was also measured in the CHW RCT as the mean number of clients at risk of pregnancy who knew they were pregnant by the end of the visit per CHW. This was higher in the intervention group (mean difference: 0.86 clients more, 95% CI: 0.59 more to 1.13 more).

Table 2 presents the GRADE evidence profile for the RCT outcomes. The four post-abortion RCTs were rated down for indirectness of the population, as we were interested in all users of pregnancy self-tests while these studies focused on a specific subset of users following abortion. They were also rated down for inconsistency, as meta-analytic results across the four studies showed both the possibility of appreciable benefit and appreciable harm. This resulted in low-certainty evidence. The CHW study was downgraded for risk of bias due to self-reporting of outcomes that may have been affected by self-report.

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Table 2. GRADE evidence profile for effectiveness review studies

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	**Study design**	**Risk of bias**	**Inconsistency**	**Indirectness**	**Imprecision**	**Other considerations**	**self-testing for pregnancy**	**clinic testing**	**Relative (95% CI)**	**Absolute (95% CI)**		
Lost to follow-up (follow up: mean 2 weeks)												
4	randomised trials	not serious [a]	not serious [b]	serious [c]	serious [d]	none	109/2737 (4.0%)	178/2730 (6.5%)	**RR 0.479** (0.155 to 1.480)	**34 fewer per 1,000** (from 55 fewer to 31 more)	⊕⊕◯◯ LOW	CRITICAL
Clients at risk of pregnancy who received antenatal counselling at visit (follow up: mean 4 months; assessed with: mean number per CHW)												
1	randomised trials	serious [e]	not serious [f]	not serious	not serious	none	N=260 Mean: 0.67	N=246 Mean: 0.28	MD **0.39 clients more** (0.14 more to 0.64 more)		⊕⊕⊕◯ MODERATE	CRITICAL
Clients at risk of pregnancy who knew they were pregnant by the end of the visit (follow up: mean 4 months; assessed with: mean number per CHW)												
1	randomised trials	serious [e]	not serious [f]	not serious	not serious	none	N=260 Mean: 0.95	N=246 Mean: 0.10	MD **0.86 clients more** (0.59 more to 1.13 more)		⊕⊕⊕◯ MODERATE	CRITICAL

CI: Confidence interval; **RR:** Risk ratio; **MD:** Mean difference; **CHW:** Community health worker

Explanations

a. Risk of bias: Blinding of health providers and participants was not possible due to the nature of the intervention, but outcome was judged to be unaffected by blinding.

b. Inconsistency: Although I-squared of 87 indicates substantial heterogeneity, we did not downgrade for inconsistency because there is likely to be true underlying inconsistency across populations, and in sensitivity analyses, all showed that there was no statistically significant difference between intervention and control groups.

c. Indirectness: Downgraded because population is individuals having medical abortions, while PICO includes both individuals desiring and not desiring pregnancy

d. Imprecision: Downgraded because 95% CI for RR includes both 1 (no effect) AND either appreciable harm (0.75) or appreciable benefit (1.25).

e. Risk of bias: Downgraded because blinding was not possible due to the nature of the intervention, and the outcome may have been affected by blinding, since it was self-reported on a survey.

f. Inconsistency: This could not be evaluated, as there is only a single study.

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For our outcome of missed ongoing pregnancy, several studies of medical abortion follow-up looked at incomplete abortion. However, we did not consider this outcome measure as "missed" pregnancies, as all were identified through the study protocols, except possibly for those lost to follow-up.

No studies measured our other outcomes of interest: self-efficacy, self-determination, autonomy, and empowerment; mental health and well-being; adverse events and social harms; and device-related issues.

Values and preferences review

Overall, 18 studies were included in the values and preferences review.[5, 13-16, 18-30] Table 3 provides descriptive data and key findings of these studies.

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Table 3. Descriptions and key findings of values and preferences studies

Study	Location	Population Description	Study design	Methods	Sample size (n)	Key values and preferences findings
End users – general						
Al-Hassan 2011	Saudi Arabia (Riyadh)	Pregnancy test kit users attending a family planning clinic	Quantitative	Cross-sectional survey	120	Reasons for using the home pregnancy test kits included: quick results (48%), convenience (25%), both speed and convenience (34%), privacy (13%), cost (8%), and accuracy (5%).
Childerhose and MacDonald, 2013	USA (online)	Pregnancy test kit users sharing their stories online	Qualitative	Stories, web logs, and video logs	99	Themes included privacy, empowerment, consumer choice, and meaning making. Home pregnancy testing "temporarily transforms these users into biomedical workers who perform […] knowledge production and medical judgment". Home pregnancy tests can be theorized as "a form of tool use that allows non-experts to produce diagnostic knowledge about their own bodies and health; and as the ongoing biopolitical work that is expected of citizens to produce healthy bodies."
Coons et al., 1990	USA (Lexington, Kentucky)	College students who had used a pregnancy test kit	Quantitative	Cross-sectional survey	131	Common reasons for using a home pregnancy test kit included: speed in obtaining results (51.9%), confidentiality (30.5%), convenience (16.8%). Cost (5.3%), accuracy (3.8%), and other (2.3%) were less significant factors.
Pike et al., 2013	UK (Manchester)	Adult female volunteers who were provided different kinds of pregnancy tests	Quantitative	Cross-sectional survey	111	After experiencing the use of different test formats, participants preferred midstream-format pregnancy tests. Reasons included hygiene concerns and the need to find a suitable container to collect the sample in when using the cassette or strip tests. Percent of respondents who liked each test (assessed as scoring 1 or 2 on a scale from 1-7 with 1 being most positive) were: strip (8.1%), cassette (3.6%), store-brand midstream visual (34.2%), branded midstream visual (31.5%), branded midstream easy-use visual (77.5%), and branded midstream digital (72.7%).
Robinson, 2020	USA	Adults who had used a pregnancy test kit when in different-sex relationships where they did not want to conceive	Qualitative	In-depth interviews	83	For women who do not want to be pregnant, home pregnancy tests have become socially and morally mandatory. The cultural script produces an assumption, impetus, and moral obligation that women must test. Women want to know if there is something inside of them that has the potential to change every aspect of their lives, including their health, their relationships, their living situation, their work, and the lives of their descendants. Women may feel pressure to home-test for pregnancy more than other health conditions because of the time constraints associated with abortion and the unavoidable

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						progression timeline of pregnancy. Women also feel more pressure to test because of the availability of the over-the-counter home pregnancy test. In addition to testing for the condition of pregnancy, the pregnancy test shapes interactions in a variety of areas of their social life including their roles, their relationships, and their responsibilities.
Ross, 2018	Scotland	Partnered adults experiencing a first pregnancy	Qualitative	Longitudinal in-depth interviews	15	Participants engaged with pregnancy tests in varying ways, with uses shaping and shaped by their experiences of early pregnancy more broadly. Particular technical characteristics of the home pregnancy test led many participants to question their interpretation of a positive result, as well as the accuracy of the test itself. Rather than addressing the unknowns of early gestation by confirming a suspected pregnancy, a positive result could thus exacerbate uncertainty.
End users – following medical abortion						
Blum et al., 2016	Vietnam (Ho Chi Minh City)	Individuals seeking medical abortion	Quantitative	Cross-sectional survey at FU	600	99% of participants found the pregnancy test acceptable in terms of time required to use at home; the rest had no opinion. When asked their preferred location for managing abortion follow-up in the future, 89% said at home with pregnancy test, while 9% said clinic and 2% said no preference.
Cameron et al., 2012	Scotland (Edinburgh)	Individuals seeking medical abortion	Quantitative	Cross-sectional survey at FU	75	All participants said they would be likely to recommend at-home pregnancy testing and phone calls as a follow-up for managing medical abortion to a friend.
Constant et al., 2017	South Africa (Cape Town)	Individuals seeking medical abortion	Quantitative	Cross-sectional survey at FU	525	Nearly all participants (98.5%) preferred self-assessment to an in-clinic follow-up appointment. Most were satisfied with their abortion (97.8%), would recommend the method to a friend (82.0%) and would want the same method, should they need another abortion (86.2%).
Dabash et al., 2016	Tunisia (Nabeul, Sousse, Ben Arous, Hamam Lif, and Tunis)	Individuals seeking medical abortion	Quantitative	Cross-sectional survey at FU	348	Most participants would consider using the pregnancy test again (97.4%) and would recommend it to a friend (97.7%) for abortion follow-up.
Harper et al., 2002	USA (17 locations)	Individuals seeking medical abortion	Quantitative	Cross-sectional survey at FU	2121	Overall, 92% of participants reported that they would be comfortable with home management. In multivariate analysis, those with higher education and while/black vs. Hispanic/Asian individuals were more likely to be comfortable.

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Hassoun et al., 2016	France (17 locations nationally)	Individuals seeking medical abortion	Quantitative	Cross-sectional survey at FU	177	Regarding acceptability of the urine pregnancy test, 39.9% of participants found it reassuring, 25.8% found it satisfactory, and 4.5% found it both reassuring and satisfactory. Smaller proportions of participants found it unsettling (12.4%), alarming (2.8%). Very small proportions of participants found it both reassuring and unsettling (1.7%), both satisfactory and alarming (0.6%), or both reassuring and alarming (0.6%); 11.8% had no opinion.
Iyengar et al., 2015	India (Rajasthan)	Individuals seeking medical abortion	Quantitative	Cross-sectional survey at FU	349	Most participants in the home-pregnancy-assessment group (334/349 [96%]) were satisfied with the method of abortion follow-up.
Iyengar et al., 2016	India (Rajasthan)	Individuals seeking medical abortion	Qualitative	In-depth interviews	20	With home-use of misoprostol, women were able to avoid inconvenience of travel, child care, and housework, as well as maintain confidentiality. The use of a low-sensitivity pregnancy test alleviated women's anxieties about retained products; they found it reassuring, and several experienced a sense of relief. They expressed that doing the test was useful, as it saved a visit to the clinic. The majority said they would prefer medical abortion involving a single visit in future.
Ngoc et al., 2014	Vietnam (Ho Chi Minh City and Hanoi)	Individuals seeking medical abortion	Quantitative	Cross-sectional survey at FU	1328	In the phone follow-up group, most participants (88.3% [606/686]) indicated that they preferred to complete their follow-up at home with a phone call. In the clinic follow-up group, fewer women (39.9% [256/642]) reported that they would opt for phone follow-up at home in the future.
Oppegaard et al., 2015	Austria (Vienna), Finland (Helsinki), Norway (Oslo), Sweden (Stockholm)	Individuals seeking medical abortion	Quantitative	Cross-sectional survey at FU	653	The majority of participants in both the self-assessment and clinic assessment groups said they would prefer self-assessment at home to routine clinical follow-up if they were to have an abortion in future (self: 272 [82%] of 330; clinic: 190 [59%] of 323, p<0·0001).
Platais et al., 2015	Moldova (Chisinau, Balti, and Drochia), Uzbekistan (Tashkent)	Individuals seeking medical abortion	Quantitative	Cross-sectional survey at FU	2400	When asked about their preference for future follow-up, most women in the phone group (76.1%) and the plurality of women in the clinic group (48.1%) preferred phone follow-up. 9.6% of the phone group and 29.1% of the clinic group preferred clinic follow-up, and the remainder had no preference.

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Shochet et al., 2017	USA (Stanford, CA), Vietnam (Ho Chi Minh City)	Individuals seeking medical abortion	Quantitative	Cross-sectional survey at FU	87	Most participants (73.6%) reported being satisfied or very satisfied with using the pregnancy tests at home, 76.7% would want to use home pregnancy tests to monitor for future fertility treatment, and 84.9% would recommend using home pregnancy tests in addition to hospital visits to a friend. 76.7% agreed using home pregnancy tests in lieu of some hospital visits would be more convenient/save time.

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There were 12 quantitative studies (all cross-sectional surveys) and four qualitative studies. For populations, six studies included general pregnancy test users or volunteers, while 12 studies followed individuals after they received a medical abortion with at-home follow-up including a home pregnancy test. No studies were identified with providers or other stakeholders.

The studies were conducted in diverse locations: USA (n=5), UK (n=3), India (n=2), Vietnam (n=2), and one each in Austria, Finland, France, Moldova, Norway, Saudi Arabia, South Africa, Sweden, Tunisia, and Uzbekistan.

Among general pregnancy test users, most studies found support for pregnancy tests. Reasons why individuals liked pregnancy tests included getting quick results, convenience, confidentiality/privacy, cost, and accuracy. One study found a preference for midstream results tests when individuals were asked to try a range of tests. This same study found that certain types of tests were less preferred due to perceived cleanliness and convenience. Finally, several qualitative studies noted that pregnancy tests shape individuals' relationships with their bodies and their social roles, relationships, and responsibilities.

Among individuals having medical abortions, across all studies, most individuals receiving home management with a pregnancy test said they would prefer this option in the future. This ranged from 76.1% (Moldova and Uzbekistan) to 98.5% (South Africa). In two trials with clinic comparison groups, home management was still the most preferred option among participants in the clinic arms. When asked, clear majorities of participants across studies said they found home management acceptable and would recommend it to a friend.

Cost review

No studies presented primary data examining cost-effectiveness, cost-utility, or cost-benefit for pregnancy self-testing.

Discussion

Pregnancy self-testing has become common practice in many high-income settings, though is still not widely available in many low-income settings globally. This systematic review is the first to our knowledge to summarize the available literature on effectiveness, values and preferences, and cost for this important intervention.

In the effectiveness review, we found low-certainty evidence from four RCTs following individuals after medical abortion showing no difference in loss to follow-up with self-testing for pregnancy at home compared to a return clinic visit. Moderate-quality evidence from one RCT showed that when CHWs were given pregnancy tests to provide to clients at home, there was a significant increase in the mean number of clients per CHW at risk of pregnancy who knew they were pregnant by the end of the visit and who received ANC counseling. There was no data on missed ongoing pregnancy; self-efficacy, self-determination, autonomy, and empowerment; mental health and well-being; adverse events and social harms; or device-related issues. Given the ubiquity of self-testing for pregnancy in many settings, it is perhaps unsurprising that this intervention has not been studied widely in a comparative way except in specific circumstances.

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Given the positive findings in these two settings (post-abortion care and CHW programs), there is evidence that wider use of home pregnancy tests could have beneficial outcomes within health systems. However, the lack of data from a wider swath of users limits the conclusions that can be drawn about public health benefits of this intervention.

Further, our effectiveness review only compared studies where individuals had access to home pregnancy tests versus facility-based testing. We excluded studies comparing groups of individuals given home pregnancy tests compared to those given nothing, or standard of care. Several trials have examined this latter question, asking whether having a home-pregnancy test on hand would cause women to have a lower threshold of suspicion for pregnancy, to test for pregnancy more often, or to identify pregnancies earlier. For example, one RCT among low-income women in the United States found that participants given home pregnancy tests did suspect pregnancy more often and test for pregnancy more often than control participants.[31] However, another larger RCT in the United States found no difference between arms in mean time at first pregnancy testing or mean gestational weeks at first positive pregnancy test.[32] In both these cases, control participants presumably had at least some access to over-the-counter home pregnancy tests, but results do speak to the potential impact of expanded access to such tests.

For values and preferences literature, 18 studies from diverse populations globally found that most individuals supported pregnancy self-testing. Reasons why individuals liked pregnancy tests included getting quick results, convenience, confidentiality/privacy, cost, and accuracy. Mid-stream urine tests were preferred. Most individuals receiving post-abortion home management with a pregnancy test said they would prefer this option in the future. Although no data from providers were identified, these positive findings from end users support broader access to pregnancy self-testing.

Our review identified no comparative cost data on pregnancy self-testing. Clearly, costs of pregnancy self-tests will vary by setting and specific product, but should generally fall within the range of other over-the-counter maternal health products. Considerations of cost should include not only the cost of the test for the end user, but also the full range of health sector costs (e.g. costs due to delayed pregnancy care), other sector costs, and productivity impacts (e.g. labor and workforce issues). Creative ways of expanding access to pregnancy self-testing within existing health care systems, such as using community health workers, would benefit from including cost and cost-effectiveness assessments.

This review had strengths and limitations. While we conducted a thorough and systematic search, we only included peer-reviewed literature; we thus may have missed grey literature. Our findings are limited by the literature available, which was heavily skewed towards literature on post-abortion care, representing only a fraction of individuals using pregnancy self-tests globally.

Conclusion

Pregnancy self-testing is acceptable and valued by end users. Ensuring universal access to pregnancy self-testing may encourage women to seek early antenatal care, which is a critical opportunity for health providers to deliver care and support to pregnant women, contributing to

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better health outcomes for women, newborn and children. Novel ways of delivering pregnancy self-test kits, including through community health worker programs, show promise on improving sexual and reproductive health outcomes, and should be considered for further study and expansion. In the current context of the COVID-19 pandemic, greater availability of self-testing should also help maintain the continuity of sexual and reproductive health services.

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Provenance and peer review: Not commissioned; externally peer reviewed.

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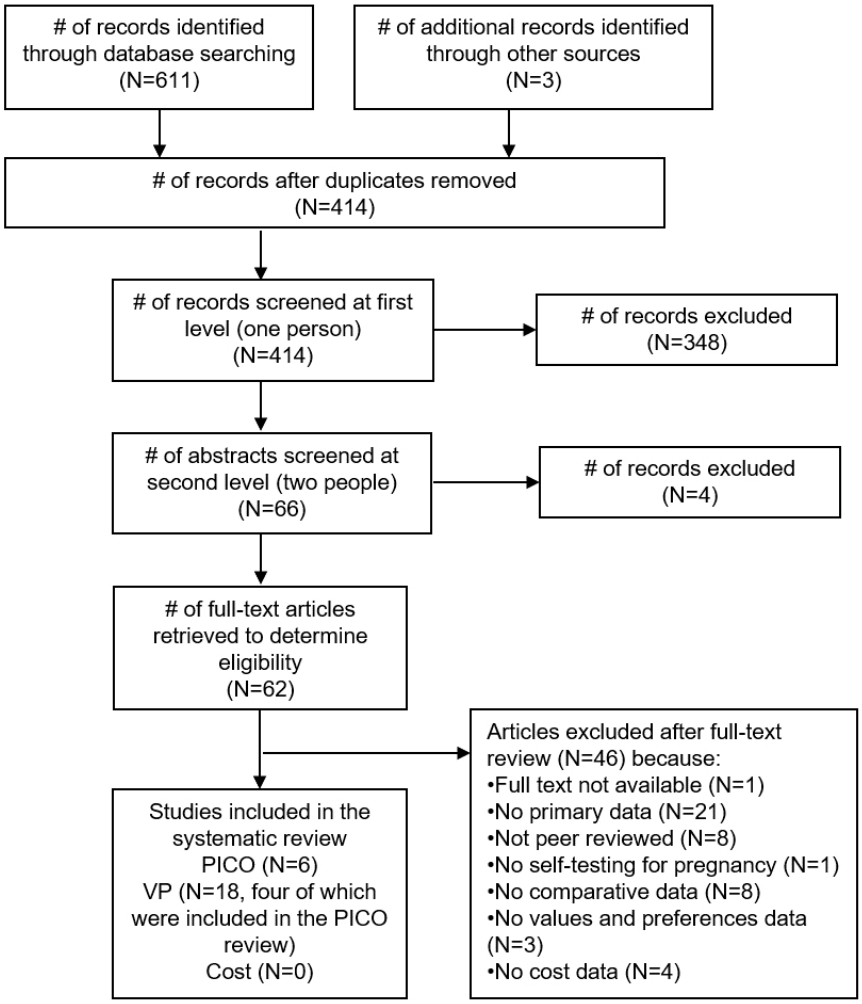


Figure 1. PRISMA flow chart showing disposition of citations through the search and screening process.

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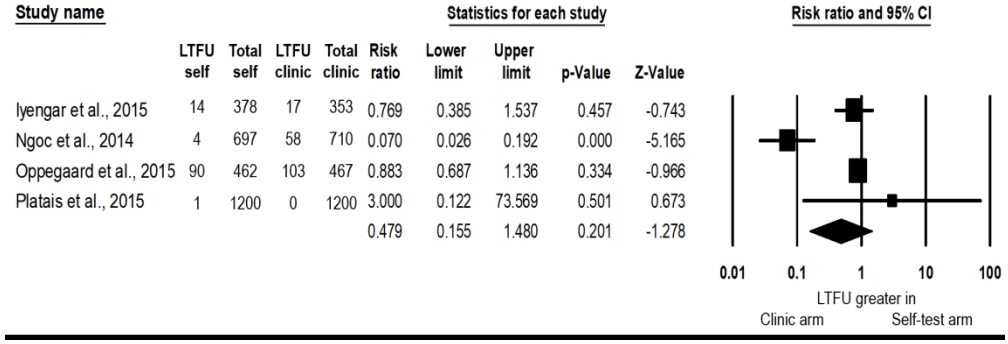


Figure 2. Meta-analysis of loss to follow-up (LTFU) among clients following medical abortion, comparing those receiving follow-up at home versus a return clinic visit

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	6
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	7

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8-9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	10
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	11-14
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	11-14
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	11-14
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	11-14
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	19-20
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	20
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	20-21
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	21

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Self-testing for pregnancy: a systematic review and meta-analysis

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Abstract

Objectives: Urine pregnancy tests are often inaccessible in low-income settings. Expanded provision of home pregnancy testing could support self-care for sexual and reproductive health and rights. We conducted a systematic review on pregnancy self-testing effectiveness, values and preferences, and cost.

Design: Systematic review and meta-analysis using the GRADE approach.

Data sources: PubMed, CINAHL, LILACS and EMBASE and four trial registries were searched through 2 November 2020.

Eligibility criteria for selecting studies: We included trials and observational studies that compared urine self-testing for pregnancy to health worker-led pregnancy testing on effectiveness outcomes, and quantitative and qualitative studies describing values and preferences of end users and health workers and costs of pregnancy self-testing.

Data extraction and synthesis: Two independent reviewers used standardized methods to search, screen, and code included studies. Risk of bias was assessed using the Cochrane Collaboration and Evidence Project tools. Meta-analysis was conducted using random effects models. Findings were summarized in GRADE evidence profiles and synthesized qualitatively.

Results: For effectiveness, four randomized trials following 5493 individuals after medical abortion showed no difference in loss to follow-up with home pregnancy self-testing compared to return clinic visits (combined risk ratio: 0.479, 95% confidence interval: 0.155 to 1.480). One additional trial of community health workers (CHWs) offering home pregnancy tests showed a significant increase in pregnancy knowledge and antenatal counseling among 506 clients. Eighteen diverse values and preferences studies found support for pregnancy self-testing because of quick results, convenience, confidentiality/privacy, cost, and accuracy. Most individuals receiving pregnancy tests for post-abortion home management preferred this option. No studies reported cost data.

Conclusion: Pregnancy self-testing is acceptable and valued by end users. Effectiveness data comes mostly from articles on post-abortion care, and cost data is lacking. Greater availability of pregnancy self-tests, including in post-abortion care and CHW programs, may lead to improved outcomes.

Keywords: Pregnancy test, self-care, systematic review, values and preferences, post-abortion care, community health workers, sexual and reproductive health

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Systematic review registration number: PROSPERO CRD42021231656

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Strengths and limitations of this study

- This systematic review and meta-analysis used a comprehensive search for articles not only on the effectiveness of pregnancy self-testing (both randomized trials and comparative observational studies) but also on values and preferences of end users and health workers, and costs.
- We used the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) approach to evaluate the strength and quality of the evidence.
- Because pregnancy self-testing is commonplace in many settings, it is perhaps unsurprising that not many studies have examined the effectiveness of this intervention on home use versus facility-based access in comparative ways, except in specific circumstances like post-abortion care and community health worker programs.
- Values and preferences data came from diverse populations around the world, suggesting high generalizability, but no comparative cost data on pregnancy self-testing were identified.

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Introduction

Urine tests for pregnancy measure the presence of human chorionic gonadotropin (hCG), and are widely used to detect pregnancy in both home and clinical settings. Urine pregnancy tests have evolved substantially since they were first developed 100 years ago.[1] The current generation of tests have close to 100% sensitivity and specificity in detecting hCG at concentrations of 25 milli-IU/ml or more, and thus are able to detect pregnancy as early as one day after a missed menstrual period.[1] However, the correct use of urine pregnancy tests varies with user characteristics. A systematic review published in 1998 identified five studies that evaluated the diagnostic efficiency of home pregnancy tests.[2] Sensitivity ranged from 0.52 to 1.0 across tests. A gradient of sensitivity was identified by user group. Sensitivity was highest in studies where urine samples obtained by the investigators were tested by volunteers (sensitivity: 0.91, 95% confidence interval [CI], 0.84-0.96) and lower in actual patients who performed the test on their own urine samples (sensitivity 0.75 [95% CI, 0.64-0.85]). Similarly, the test effectiveness score (discriminatory power, where a score of 1.0 or less indicates poor distinctions between pregnant and nonpregnant individuals, and higher scores imply greater effectiveness) was 2.75 (95% CI, 2.3-3.2) for studies where subjects were volunteers but 0.82 (95% CI, 0.4-1.2) for studies with actual patients.

Providing pregnancy tests for home use may have a range of benefits for different populations. In Madagascar, randomized trial data has shown that providing pregnancy tests to community health workers for home distribution can increase both engagement in antenatal care services[3] and contraceptive services, since a negative pregnancy test is necessary before initiating some contraceptive methods.[4] Home pregnancy tests have been shown to be an acceptable and feasible option for follow-up among couples undergoing assisted reproduction.[5] There is also evidence supporting the efficacy, safety, and acceptability of urine pregnancy tests to confirm the effectiveness of a medical abortion instead of an ultrasound.[6, 7] While urine pregnancy self-tests are available over-the-counter in many high- and middle-income settings, in many low-income settings, they may be financially inaccessible to most people outside of public health services, or unavailable altogether, leading individuals with the sole option of health facility-based blood tests to confirm pregnancy. For example, most countries in the Eastern Mediterranean Region have pregnancy self-testing widely available in private pharmacies, particularly in urban settings, and mainly used by the upper socio-economic class due cost and knowledge.[8]

Many people in resource-constrained settings are not able to decide if, how many, and when to have children; increased access to self-care interventions, such as pregnancy self-tests, could support their health decision-making. More widespread efforts to provide pregnancy self-tests that can be used at home or another preferred location could support increased autonomy of individuals as well as support multiple programmatic settings to advance sexual and reproductive health and rights.

We sought to review the literature on home use of pregnancy self-testing as an additional option to facility-based testing. We conducted this systematic review in the context of expanding the evidence base of the 2019 WHO's normative guidance on self-care interventions[9] to include new considerations related to sexual and reproductive health. This review was also conducted in

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the context of the COVID-19 pandemic which continues to result in significant disruptions of essential sexual and reproductive health services. The restrictive measures taken to prevent the spread of COVID-19, particularly lockdowns, have resulted in the need for increased availability and accessibility to this self-care intervention.

Methods

This review addressed the question: should self-testing for pregnancy be available as an additional option to health facility-based testing? We reviewed the extant literature in three areas relevant to answering this question: effectiveness of the intervention, values and preferences of end users and health workers, and cost information. The review followed PRISMA guidelines,[10] and the protocol was published on PROSPERO (CRD42021231656). Ethical approval was not required for this systematic review, since all data came from published articles.

Effectiveness review

The effectiveness review was designed according to the population-intervention-comparison-outcomes (PICO) format as follows:

Population: Individuals seeking pregnancy testing
Intervention: Urine self-testing for pregnancy
Comparison: Health worker-led testing for pregnancy (health facility or community clinic with either urine and/or serum test for pregnancy)
Outcomes:
(1) Missed ongoing pregnancy
(2) Appropriate clinical follow-up (counseling, antenatal care visit(s), contraceptive services, abortion services, etc.)
(3) Gestational age at pregnancy awareness (knowledge of pregnancy) and at presentation for antenatal care or abortion
(4) Self-efficacy, self-determination, autonomy, and empowerment
(5) Mental health and well-being (anxiety, stress, self-harm)
(6) Adverse events and social harms (including stigma, discrimination, coercion, violence [including intimate-partner violence, violence from family members or community members, etc.] and breaches of confidentiality), and whether these harms were corrected/had redress available
(7) Device-related issues (e.g. test failure, problems with manufacturing, packaging, labelling, or instructions for use)

All results that were compatible with each outcome domain in each study were extracted.

Inclusion criteria

To be included in the review, an article had to meet the following criteria:

1) Study design that compared urine self-testing for pregnancy to health worker-led testing for pregnancy (health facility or community services with either urine and/or serum test

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for pregnancy). This included both randomized controlled trials, non-randomized controlled trials and comparative observational studies (including prospective controlled cohort studies, cross-sectional studies, controlled before-after studies and interrupted time series) that compare individuals who received the intervention to those who did not.
2) Measured one or more of the outcomes listed above.
3) Published in a peer-reviewed journal.

No restrictions were placed based on location of the intervention. No language restrictions were used on the search. Articles in English, French, Spanish, and Chinese were coded directly; articles in other languages were translated.

Search strategy

The following electronic databases were searched through the search date of 2 November 2020: PubMed, CINAHL, LILACS and EMBASE. We searched for ongoing randomized controlled trials (RCTs) through clinicaltrials.gov, the WHO ICTRP, PACTR, and the Australian New Zealand Clinical Trials Registry. Secondary reference searching was also conducted on all studies included in the review. Further, selected experts in the field were contacted to identify additional articles not identified through other search methods. The full search strategy was developed for Pubmed and adapted for entry into all computer databases (Appendix A). These search terms were used for the main systematic review (PICO question), for the values and preferences review, and for the cost review.

Screening abstracts

Titles, abstracts, citation information, and descriptor terms of citations identified through the search strategy were screened by a member of the senior study staff. Full text articles were obtained of all selected abstracts, and two independent reviewers assessed all full-text articles for eligibility to determine final study selection. Differences were resolved through consensus.

Data extraction and management

Data were extracted independently by two reviewers using standardized data extraction forms. Differences in data extraction were resolved through consensus, and referral to a senior study team member from WHO when necessary.

The following information was gathered from each included study:

- Study identification: Author(s); type of citation; year of publication
- Study description: Study objectives; location; population characteristics; type of urine pregnancy test; description of self-test access; study design; sample size; follow-up periods and loss to follow-up
- Outcomes: Analytic approach; outcome measures; comparison groups; effect sizes; confidence intervals; significance levels; conclusions; limitations

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For randomized trials, risk of bias was assessed using the Cochrane Collaboration's tool for assessing risk of bias.[11] For non-randomized trials but comparative studies, study rigor was assessed using the Evidence Project 8-item checklist for intervention evaluations.[12]

Data analysis

Data were analyzed according to coding categories and outcomes. Where multiple studies reported the same outcome, meta-analysis was conducted using random-effects models to combine risk ratios with the program Comprehensive Meta-Analysis (CMA).[13] Risk ratios were used directly when provided, and were calculated from the number of events and total sample size in the intervention and comparison groups when not reported. Heterogeneity was calculated using the I-squared statistic.

All analyses were stratified by the following categories/subgroups, where possible:

- Pregnancy self-test point of access (e.g. pharmacy, online, community health worker (CHW) distribution)
- Type of pregnancy test
- Population (e.g. age, marital status, rural/urban)
- Vulnerabilities (e.g. obesity, socio-economic status, poverty, disability, literacy/educational level)
- Gestational age
- High-income versus low or middle-income countries

Data were summarized in GRADE Evidence Profile tables using GRADEPro. Where RCT data were available for a given outcome, we prioritized that over observational data.

Values and preferences review

The same search terms and screening process were used to identify studies to be included in the values and preferences review. Studies were included in this review if they presented primary data examining preferences of individuals regarding urine self-testing for pregnancy. We focused on studies examining the values and preferences of women and adolescent girls who were self-testing for pregnancy or individuals who were potential candidates for such self-testing, but we also included studies examining the values and preferences of health workers and other stakeholders. In this section, we also considered issues related to ability to access (by age, gender, or other factors. These could include legal restrictions around who can access e.g. by age, requirement to be married to purchase, etc. These could also include practical barriers, e.g. is it harder for adolescents to access self-tests because they are not allowed out on their own, they have no money, etc.), informed decision-making, coercion, confidentiality, self-determination, health decision-making to terminate or maintain pregnancy (and discussion with partner if appropriate), and seeking redress. These studies could be qualitative or quantitative in nature, but had to present primary data collection – think pieces and review articles were not included. Values and preferences literature was summarized qualitatively and organized by study design and methodology, location, and population.

Cost review

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The same search terms and screening process were used to identify studies to be included in the cost review. Studies would have been included in this review if they presented primary data comparing costing, cost-effectiveness, cost-utility, or cost-benefit of the intervention and comparison listed in the PICOs above, or if they presented cost-effectiveness of the intervention as it relates to the PICO outcomes listed above. If cost literature had been found, we would have summarized it qualitatively. We planned to classify cost literature into four categories (health sector costs, other sector costs, patient/family costs, and productivity impacts) and within each category would have organized by study design/methodology, location, and population.

Patient and public involvement

Several of the authors are current or previous users of pregnancy self-tests. Feedback was also received from the WHO patient safety working group. Patients were involved in a global survey of values and preferences conducted to inform the WHO guideline on self-care interventions; they thus play a significant role in the overall recommendation informed by this review.

Results

Our search yielded 414 unique references, of which 62 were retained for full-text review (Figure 1). Ultimately, we identified six that met the inclusion criteria for the effectiveness review, 18 values and preferences studies, and no cost studies.

Figure 1. PRISMA flow chart showing disposition of citations through the search and screening process

Effectiveness review

Overall, six studies met the inclusion criteria for the effectiveness review. This included five RCTs[3, 14-17] and 1 observational study.[18] Table 1 presents descriptive data from the 5 RCTs. Four of the RCTs, conducted in a diverse range of countries (India, Vietnam, Austria, Finland, Norway, Sweden, Moldova, and Uzbekistan) were conducted among 5493 individuals receiving medical abortion.[14-17] These RCTs randomized clients to abortion follow-up with home pregnancy testing and a phone call, versus abortion follow-up with the traditional clinic visit, usually with ultrasound confirmation of successful termination. The fifth RCT, conducted in Madagascar, randomized CHWs to receive pregnancy tests to use with their clients versus the standard of care, which the authors said had historically been pregnancy testing available only at clinics.[3]

Table 1. Description of RCTs included in the effectiveness review

Study	Location	Sample size	Intervention	Comparator	Outcomes

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Women who had a medical abortion					
Iyengar et al., 2015	India (Rajasthan: urban and rural)	N=731	Abortion follow-up with home pregnancy testing and phone call	Abortion follow-up with clinic visit	2. Appropriate clinical follow-up
Ngoc et al., 2014	Vietnam (Ho Chi Minh City and Hanoi)	N=1433			
Oppegaard et al., 2015	Austria (Vienna), Finland (Helsinki), Norway (Oslo), Sweden (Stockholm)	N=929			
Platais et al., 2015	Moldova (Chisinau, Balti, and Drochia), Uzbekistan (Tashkent)	N=2400			
Community health workers and their clients					
Comfort et al., 2019	Madagascar (Aloatra Mangoro, Atsinanana and Analanjirofo)	N=506 CHWs and their clients	Providing CHWs with pregnancy tests to give to clients	Pregnancy testing available at clinics	2. Appropriate clinical follow-up 3. Gestational age at pregnancy awareness (knowledge of pregnancy) and at presentation for antenatal care or abortion

The RCTs provided data for two of our PICO outcomes: 2) Appropriate clinical follow-up, and 3) Gestational age at pregnancy awareness (knowledge of pregnancy) and at presentation for antenatal care or abortion. The non-randomized observational study provided data only for the same outcome under appropriate clinical follow-up, so per our protocol, we present only the RCT data here.

Appropriate clinical follow-up was assessed in the four post-abortion RCTs as loss to follow-up, meaning the client did not return for their follow-up visit or was not able to be contacted by phone. While one RCT from Vietnam found dramatically improved follow-up in the pregnancy self-testing arm, with only 0.6% of individuals lost compared with 8.1% lost in the comparison arm [15], the other three RCTs found no significant differences between study arms. In meta-analysis, there was no significant difference between study arms in loss to follow-up (pooled risk ratio: 0.479, 95% confidence interval 0.155 to 1.480) (Figure 2). Heterogeneity was substantial, with an I-squared of 87. Stratification by high- vs. low- and middle-income settings did not yield meaningful differences; no further stratifications were available.

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Figure 2. Meta-analysis of loss to follow-up (LTFU) among clients following medical abortion, comparing those receiving follow-up at home versus a return clinic visit

Appropriate clinical follow-up was assessed in the CHW RCT as the mean number of clients at risk of pregnancy who received antenatal counselling at their CHW visit per CHW. This was higher in the intervention group (mean difference: 0.39 clients more, 95% CI: 0.14 more to 0.64 more).

Gestational age at pregnancy awareness (knowledge of pregnancy) and at presentation for antenatal care or abortion was also measured in the CHW RCT as the mean number of clients at risk of pregnancy who knew they were pregnant by the end of the visit per CHW. This was higher in the intervention group (mean difference: 0.86 clients more, 95% CI: 0.59 more to 1.13 more).

Table 2 presents the GRADE evidence profile for the RCT outcomes. The four post-abortion RCTs were rated down for indirectness of the population, as we were interested in all users of pregnancy self-tests while these studies focused on a specific subset of users following abortion. They were also rated down for inconsistency, as meta-analytic results across the four studies showed both the possibility of appreciable benefit and appreciable harm. This resulted in low-certainty evidence. The CHW study was downgraded for risk of bias due to self-reporting of outcomes that may have been affected by self-report.

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Table 2. GRADE evidence profile for effectiveness review studies

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	**Study design**	**Risk of bias**	**Inconsistency**	**Indirectness**	**Imprecision**	**Other considerations**	**self-testing for pregnancy**	**clinic testing**	**Relative (95% CI)**	**Absolute (95% CI)**		
Lost to follow-up (follow up: mean 2 weeks)												
4	randomised trials	not serious [a]	not serious [b]	serious [c]	serious [d]	none	109/2737 (4.0%)	178/2730 (6.5%)	**RR 0.479** (0.155 to 1.480)	**34 fewer per 1,000** (from 55 fewer to 31 more)	⊕⊕◯◯ LOW	CRITICAL
Clients at risk of pregnancy who received antenatal counselling at visit (follow up: mean 4 months; assessed with: mean number per CHW)												
1	randomised trials	serious [e]	not serious [f]	not serious	not serious	none	N=260 Mean: 0.67	N=246 Mean: 0.28	MD **0.39 clients more** (0.14 more to 0.64 more)		⊕⊕⊕◯ MODERATE	CRITICAL
Clients at risk of pregnancy who knew they were pregnant by the end of the visit (follow up: mean 4 months; assessed with: mean number per CHW)												
1	randomised trials	serious [e]	not serious [f]	not serious	not serious	none	N=260 Mean: 0.95	N=246 Mean: 0.10	MD **0.86 clients more** (0.59 more to 1.13 more)		⊕⊕⊕◯ MODERATE	CRITICAL

CI: Confidence interval; **RR:** Risk ratio; **MD:** Mean difference; **CHW:** Community health worker

Explanations

a. Risk of bias: Blinding of health workers and participants was not possible due to the nature of the intervention, but outcome was judged to be unaffected by blinding.

b. Inconsistency: Although I-squared of 87 indicates substantial heterogeneity, we did not downgrade for inconsistency because there is likely to be true underlying inconsistency across populations, and in sensitivity analyses, all showed that there was no statistically significant difference between intervention and control groups.

c. Indirectness: Downgraded because population is individuals having medical abortions, while PICO includes both individuals desiring and not desiring pregnancy

d. Imprecision: Downgraded because 95% CI for RR includes both 1 (no effect) AND either appreciable harm (0.75) or appreciable benefit (1.25).

e. Risk of bias: Downgraded because blinding was not possible due to the nature of the intervention, and the outcome may have been affected by blinding, since it was self-reported on a survey.

f. Inconsistency: This could not be evaluated, as there is only a single study.

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For our outcome of missed ongoing pregnancy, several studies of medical abortion follow-up looked at incomplete abortion. However, we did not consider this outcome measure as "missed" pregnancies, as all were identified through the study protocols, except possibly for those lost to follow-up.

No studies measured our other outcomes of interest: self-efficacy, self-determination, autonomy, and empowerment; mental health and well-being; adverse events and social harms; and device-related issues.

Values and preferences review

Overall, 18 studies were included in the values and preferences review.[5, 14-17, 19-31] Table 3 provides descriptive data and key findings of these studies.

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Table 3. Descriptions and key findings of values and preferences studies

Study Location	Population Description	Study design: Methods, Sample size	Key values and preferences findings
End users - general			
Al-Hassan, 2011 Saudi Arabia (Riyadh)	Pregnancy test kit users attending a family planning clinic	Quantitative: Cross-sectional survey (CSS), n=120	Reasons for using the home pregnancy test kits included: quick results (48%), convenience (25%), both speed and convenience (34%), privacy (13%), cost (8%), and accuracy (5%).
Childerhose and MacDonald, 2013 USA (online)	Pregnancy test kit users sharing their stories online	Qualitative: Stories, web logs, and video logs, n=99	Themes included privacy, empowerment, consumer choice, and meaning-making. Home pregnancy testing "temporarily transforms these users into biomedical workers who perform […] knowledge production and medical judgment". Home pregnancy tests can be theorized as "a form of tool use that allows non-experts to produce diagnostic knowledge about their own bodies and health; and as the ongoing biopolitical work that is expected of citizens to produce healthy bodies."
Coons et al., 1990 USA (Lexington, Kentucky)	College students who had used a pregnancy test kit	Quantitative: CSS, n=131	Common reasons for using a home pregnancy test kit included: speed in obtaining results (51.9%), confidentiality (30.5%), convenience (16.8%). Cost (5.3%), accuracy (3.8%), and other (2.3%) were less significant factors.
Pike et al., 2013 UK (Manchester)	Adult female volunteers who were provided different kinds of pregnancy tests	Quantitative: CSS, n=111	After experiencing the use of different test formats, participants preferred midstream-format pregnancy tests. Reasons included hygiene concerns and the need to find a suitable container to collect the sample in when using the cassette or strip tests. Percent of respondents who liked each test (assessed as scoring 1 or 2 on a scale from 1-7 with 1 being most positive) were: strip (8.1%), cassette (3.6%), store-brand midstream visual (34.2%), branded midstream visual (31.5%), branded midstream easy-use visual (77.5%), and branded midstream digital (72.7%).
Robinson, 2020 USA	Adults who had used a pregnancy test kit when in different-sex relationships where they did not want to conceive	Qualitative: In-depth interviews (IDIs), n=83	For women who do not want to be pregnant, home pregnancy tests have become socially and morally mandatory. The cultural script produces an assumption, impetus, and moral obligation that women must test. Women want to know if there is something inside of them that has the potential to change every aspect of their lives, including their health, their relationships, their living situation, their work, and the lives of their descendants. Women may feel pressure to home-test for pregnancy more than other health conditions because of the time constraints associated with abortion and the unavoidable progression timeline of pregnancy. Women also feel more pressure to test because of the availability of the over-the-counter home pregnancy test. In addition to testing for the condition of pregnancy, the pregnancy test shapes interactions in a variety of areas of their social life including their roles, their relationships, and their responsibilities.
Ross, 2018 Scotland	Partnered adults experiencing a first pregnancy	Qualitative: Longitudinal IDIs, n=15	Participants engaged with pregnancy tests in varying ways, with uses shaping and shaped by their experiences of early pregnancy more broadly. Particular technical characteristics of the home pregnancy test led many participants to question their interpretation of a positive result, as well as the accuracy of the test itself. Rather than addressing the unknowns of early gestation by confirming a suspected pregnancy, a positive result could thus exacerbate uncertainty.
End users – following medical abortion			
Blum et al., 2016 Vietnam (Ho Chi Minh City)	Individuals seeking medical abortion	Quantitative: CSS at follow-up (FU), n=600	99% of participants found the pregnancy test acceptable in terms of time required to use at home; the rest had no opinion. When asked their preferred location for managing abortion follow-up in the future, 89% said at home with pregnancy test, while 9% said clinic and 2% said no preference.

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Cameron et al., 2012 Scotland (Edinburgh)	Individuals seeking medical abortion	Quantitative: CSS at FU, n=75	All participants said they would be likely to recommend at-home pregnancy testing and phone calls as a follow-up for managing medical abortion to a friend.
Constant et al., 2017 South Africa (Cape Town)	Individuals seeking medical abortion	Quantitative: CSS at FU, n=525	Nearly all participants (98.5%) preferred self-assessment to an in-clinic follow-up appointment. Most were satisfied with their abortion (97.8%), would recommend the method to a friend (82.0%) and would want the same method, should they need another abortion (86.2%).
Dabash et al., 2016 Tunisia (Nabeul, Sousse, Ben Arous, Hamam Lif, and Tunis)	Individuals seeking medical abortion	Quantitative: CSS at FU, n=348	Most participants would consider using the pregnancy test again (97.4%) and would recommend it to a friend (97.7%) for abortion follow-up.
Harper et al., 2002 USA (17 locations)	Individuals seeking medical abortion	Quantitative: CSS at FU, n=2121	Overall, 92% of participants reported that they would be comfortable with home management. In multivariate analysis, those with higher education and white/black vs. Hispanic/Asian individuals were more likely to be comfortable.
Hassoun et al., 2016 France (17 locations nationally)	Individuals seeking medical abortion	Quantitative: CSS at FU, n=177	Regarding acceptability of the urine pregnancy test, 39.9% of participants found it reassuring, 25.8% found it satisfactory, and 4.5% found it both reassuring and satisfactory. Smaller proportions of participants found it unsettling (12.4%), alarming (2.8%). Very small proportions of participants found it both reassuring and unsettling (1.7%), both satisfactory and alarming (0.6%), or both reassuring and alarming (0.6%); 11.8% had no opinion.
Iyengar et al., 2015 India (Rajasthan)	Individuals seeking medical abortion	Quantitative: CSS at FU, n=349	Most participants in the home-pregnancy-assessment group (334/349 [96%]) were satisfied with the method of abortion follow-up.
Iyengar et al., 2016 India (Rajasthan)	Individuals seeking medical abortion	Qualitative: IDIs, n=20	With home-use of misoprostol, women were able to avoid inconvenience of travel, child care, and housework, as well as maintain confidentiality. The use of a low-sensitivity pregnancy test alleviated women's anxieties about retained products; they found it reassuring, and several experienced a sense of relief. They expressed that doing the test was useful, as it saved a visit to the clinic. The majority said they would prefer medical abortion involving a single visit in future.
Ngoc et al., 2014 Vietnam (Ho Chi Minh City and Hanoi)	Individuals seeking medical abortion	Quantitative: CSS at FU, n=1328	In the phone follow-up group, most participants (88.3% [606/686]) indicated that they preferred to complete their follow-up at home with a phone call. In the clinic follow-up group, fewer women (39.9% [256/642]) reported that they would opt for phone follow-up at home in the future.
Oppegaard et al., 2015 Austria (Vienna), Finland (Helsinki), Norway (Oslo), Sweden (Stockholm)	Individuals seeking medical abortion	Quantitative: CSS at FU, n=653	The majority of participants in both the self-assessment and clinic assessment groups said they would prefer self-assessment at home to routine clinical follow-up if they were to have an abortion in future (self: 272 [82%] of 330; clinic: 190 [59%] of 323, p<0·0001).
Platais et al., 2015 Moldova (Chisinau, Balti, and Drochia), Uzbekistan (Tashkent)	Individuals seeking medical abortion	Quantitative: CSS at FU, n=2400	When asked about their preference for future follow-up, most women in the phone group (76.1%) and the plurality of women in the clinic group (48.1%) preferred phone follow-up. 9.6% of the phone group and 29.1% of the clinic group preferred clinic follow-up, and the remainder had no preference.
Shochet et al., 2017 USA (Stanford, CA), Vietnam (Ho Chi Minh City)	Individuals seeking medical abortion	Quantitative: CSS at FU, n=87	Most participants (73.6%) reported being satisfied or very satisfied with using the pregnancy tests at home, 76.7% would want to use home pregnancy tests to monitor for future fertility treatment, and 84.9% would recommend using home pregnancy tests in addition to hospital visits to a friend. 76.7% agreed using home pregnancy tests in lieu of some hospital visits would be more convenient/save time.

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There were 12 quantitative studies (all cross-sectional surveys) and four qualitative studies. For populations, six studies included general pregnancy test users or volunteers, while 12 studies followed individuals after they received a medical abortion with at-home follow-up including a home pregnancy test. No studies were identified with health workers or other stakeholders.

The studies were conducted in diverse locations: USA (n=5), UK (n=3), India (n=2), Vietnam (n=2), and one each in Austria, Finland, France, Moldova, Norway, Saudi Arabia, South Africa, Sweden, Tunisia, and Uzbekistan.

Among general pregnancy test users, most studies found support for pregnancy tests. Reasons why individuals liked pregnancy tests included getting quick results, convenience, confidentiality/privacy, cost, and accuracy. One study found a preference for midstream results tests when individuals were asked to try a range of tests [29]. This same study found that certain types of tests were less preferred due to perceived cleanliness and convenience.

Among individuals having medical abortions, across all studies, most individuals receiving home management with a pregnancy test said they would prefer this option in the future. This ranged from 76.1% (Moldova and Uzbekistan) to 98.5% (South Africa). In two trials with clinic comparison groups, home management was still the most preferred option among participants in the clinic arms. When asked, clear majorities of participants across studies said they found home management acceptable and would recommend it to a friend.

While most values and preferences studies focused on acceptability, ease of use, and reasons for preferring pregnancy self-testing over other options, few studies also presented findings on our other areas of interest: ability to access, informed decision-making, coercion, confidentiality, self-determination, health decision-making, and seeking redress. One study from the USA found themes of privacy, empowerment, consumer choice, and meaning-making among women using pregnancy tests [22]. Other qualitative studies from the USA and Scotland noted that pregnancy tests shape individuals' relationships with their bodies and their social roles, relationships, and responsibilities [30-31]; one theorized that testing is seen as socially and morally mandatory as "the cultural script produces an assumption, impetus, and moral obligation that women must test" to know what is inside their bodies and how it may shape their lives [30]. Finally, one study from India found that managing abortion at home meant that women were able to avoid the inconvenience of travel, childcare, and housework, as well as maintain confidentiality [28]. The pregnancy test alleviated women's anxieties: they found it reassuring, and several experienced a sense of relief.

Cost review

No studies presented primary data examining cost-effectiveness, cost-utility, or cost-benefit for pregnancy self-testing.

Discussion

Pregnancy self-testing has become common practice in many high- and middle-income settings, though is still not widely available in many low-income settings globally. Increased access and availability of pregnancy self-tests for home use can help confirm a pregnancy and engage with

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the health system for improved reproductive, maternal and newborn health outcomes. Pregnancy self-tests are also an important self-care intervention that increase agency and autonomy in sexual and reproductive decision-making. This systematic review is the first to our knowledge to summarize the available literature on effectiveness, values and preferences, and cost for this important intervention.

In the effectiveness review, we found low-certainty evidence from four RCTs following individuals after medical abortion showing no difference in loss to follow-up with self-testing for pregnancy at home compared to a return clinic visit. Moderate-quality evidence from one RCT showed that when CHWs were given pregnancy tests to provide to clients at home, there was a significant increase in the mean number of clients per CHW at risk of pregnancy who knew they were pregnant by the end of the visit and who received ANC counseling. There was no data on missed ongoing pregnancy; self-efficacy, self-determination, autonomy, and empowerment; mental health and well-being; adverse events and social harms; or device-related issues. Given the ubiquity of self-testing for pregnancy in many settings, it is perhaps unsurprising that this intervention has not been studied widely in a comparative way except in specific circumstances. Given the positive findings in these two settings (post-abortion care and CHW programs), there is evidence that wider use of home pregnancy tests could have beneficial outcomes within health systems. However, fact that almost all data came from post-abortion care settings limits the generalizability of our findings and the conclusions that can be drawn about public health benefits of this intervention. In addition, the certainty of evidence in the context of post-abortion care was limited by the small number of studies combined in meta-analysis and high heterogeneity.

Further, our effectiveness review only compared studies where individuals had access to home pregnancy tests versus facility-based testing. We excluded studies comparing groups of individuals given home pregnancy tests compared to those given nothing, or standard of care. Several trials have examined this latter question, asking whether having a home-pregnancy test on hand would cause women to have a lower threshold of suspicion for pregnancy, to test for pregnancy more often, or to identify pregnancies earlier. For example, one RCT among low-income women in the United States found that participants given home pregnancy tests did suspect pregnancy more often and test for pregnancy more often than control participants.[32] However, another larger RCT in the United States found no difference between arms in mean time at first pregnancy testing or mean gestational weeks at first positive pregnancy test.[33] In both these cases, control participants presumably had at least some access to over-the-counter home pregnancy tests, but results do speak to the potential impact of expanded access to such tests.

For values and preferences literature, 18 studies from diverse populations globally found that most individuals supported pregnancy self-testing. Reasons why individuals liked pregnancy tests included getting quick results, convenience, confidentiality/privacy, cost, and accuracy. Mid-stream urine tests were preferred. Most individuals receiving post-abortion home management with a pregnancy test said they would prefer this option in the future. Although no data from health workers were identified, these positive findings from end users support broader access to pregnancy self-testing.

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Our review identified no comparative cost data on pregnancy self-testing. Clearly, costs of pregnancy self-tests will vary by setting and specific product, but should generally fall within the range of other over-the-counter maternal health products. Considerations of cost should include not only the cost of the test for the end user, but also the full range of health sector costs (e.g. costs due to delayed pregnancy care), other sector costs, and productivity impacts (e.g. labor and workforce issues). Creative ways of expanding access to pregnancy self-testing within existing health care systems, such as using community health workers, would benefit from including cost and cost-effectiveness assessments.

This review had strengths and limitations. While we conducted a thorough and systematic search, we only included peer-reviewed literature; we thus may have missed grey literature. Strengths also included our double extraction of data and inclusion of not only effectiveness literature, but also literature on values and preferences and costs. However, our findings across these three areas are limited by the literature available,. We identified no cost data, and effectiveness data was heavily skewed towards literature on post-abortion care, representing only a fraction of individuals using pregnancy self-tests globally. Further research on the effectiveness (especially on outcomes for which we found no comparative data, i.e. missed ongoing pregnancy; self-efficacy, self-determination, autonomy, and empowerment; mental health and well-being; adverse events and social harms; and device-related issues), values and preferences, and costs of pregnancy self-testing is needed particularly in low-resource settings and in contexts where people would benefit from greater accessibility.

Ultimately, the WHO Consolidated guideline on self-care interventions for health and well-being published in 2021 included the following recommendation: "WHO recommends making self-testing for pregnancy available as an additional option to health worker-led testing for pregnancy, for individuals seeking pregnancy testing. (Strong recommendation; very low certainty evidence)".[34]

Conclusion

Pregnancy self-testing is acceptable and valued by end users. Ensuring universal access to pregnancy self-testing may encourage more women and girls to seek early antenatal care, which is a critical opportunity for health workers to deliver care and support during pregnancy, thus contributing to better health outcomes for women, newborn and children. Novel ways of delivering pregnancy self-tests, including through community health worker programs, show promise on improving sexual and reproductive health and rights outcomes, and should be considered for further study and expansion. In the current context of the COVID-19 pandemic, greater availability of pregnancy self-testing can also help maintain the continuity and quality of sexual and reproductive health services.

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Contributions: MN conceptualized the study following discussion with KG and other WHO Regional Advisors from the Eastern Mediterranean and Africa Regions. CK and PTY designed the protocol, with feedback from MN. PTY ran the database search and oversaw the search, screening, full text review, and data extraction process. CK and PTY drafted the manuscript. All authors reviewed the draft, provided critical review, and read and approved the final manuscript. The corresponding author, as guarantor, accepts full responsibility for the finished article has access to any data and controlled the decision to publish. The corresponding author attests that all listed authors meet the authorship criteria and that no others meeting the criteria have been omitted.

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Provenance and peer review: Not commissioned; externally peer reviewed.

Data availability statement: All data relevant to the study are included in the article or uploaded as supplementary information. Extracted data are available on request to the corresponding author.

Transparency declaration: The corresponding author, as guarantor, affirms that the manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned (and, if relevant, registered) have been explained.

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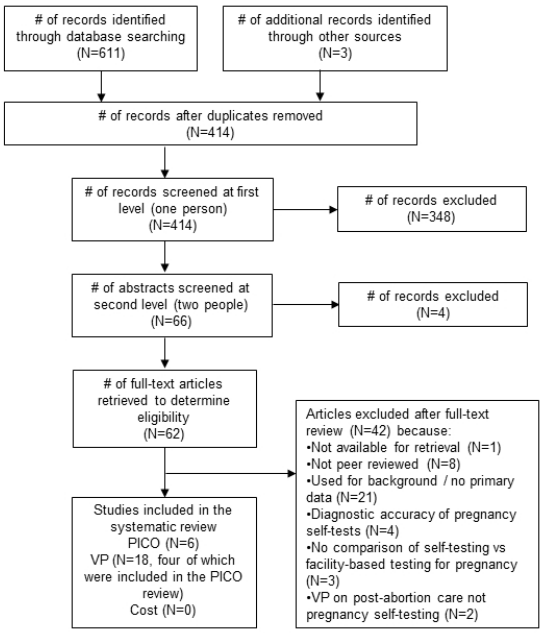


Figure 1. PRISMA flow chart showing disposition of citations through the search and screening process

254x190mm (96 x 96 DPI)

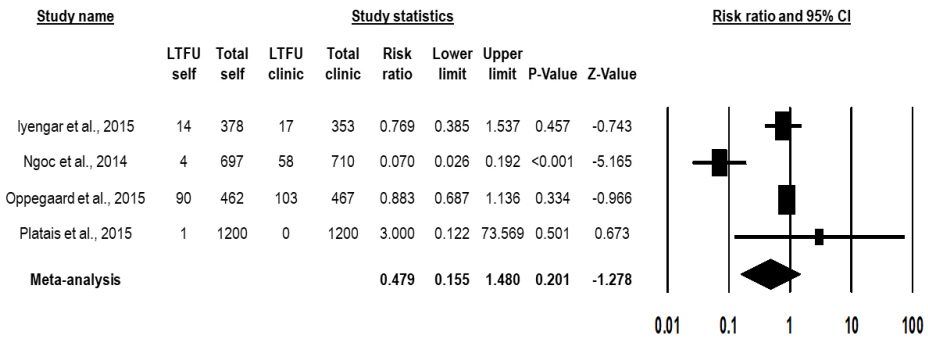


Figure 2. Meta-analysis of loss to follow-up (LTFU) among clients following medical abortion, comparing those receiving follow-up at home versus a return clinic visit

338x190mm (96 x 96 DPI)

Self-testing for pregnancy: a systematic review and meta-analysis
Appendix A. Search Strategy

Search date: 02 Nov 2020

Pubmed
("pregnancy tests" [Mesh] OR "Pregnancy Tests, Immunologic" [Mesh] OR "pregnancy test" [tiab] OR "pregnancy diagnosis" [tiab])

AND

("self care"[Mesh] OR "self-care"[tiab] OR "self-test"[tiab] OR "self-evaluation"[tiab] OR "self-evaluate"[tiab] OR "self-management"[tiab] OR "self-monitor"[tiab] OR "self-manage"[tiab] OR "self-monitored"[tiab] OR "self-managed"[tiab] OR "home"[tiab])

CINAHL
AB: "pregnancy test" OR "pregnancy diagnosis"

AND

AB: "self care" OR "self-test" OR "self-evaluation" OR "self-evaluate" OR "self-management" OR "self-manage" OR "self-monitor" OR "self-monitored" OR "home"

LILACS
"pregnancy test" OR "pregnancy diagnosis"

AND

"self care" OR "self-test" OR "self-evaluation" OR "self-evaluate" OR "self-management" OR "self-manage" OR "self-monitor" OR "self-monitored" OR "home"

EMBASE
('pregnancy test':ti,ab,kw OR 'pregnancy diagnosis':ti,ab,kw) AND ('self care':ti,ab,kw OR 'self-test':ti,ab,kw OR 'self-evaluation':ti,ab,kw OR 'self-evaluate':ti,ab,kw OR 'self-management':ti,ab,kw OR 'self-manage':ti,ab,kw OR 'self-monitor':ti,ab,kw OR 'self-monitored':ti,ab,kw OR 'home':ti,ab,kw)

clinicaltrials.gov
"pregnancy test" OR "pregnancy diagnosis"

WHO ICTRP
"pregnancy test" OR "pregnancy diagnosis"

PACTR
"pregnancy test" OR "pregnancy diagnosis"

Australian New Zealand Clinical Trials Registry
"pregnancy test" OR "pregnancy diagnosis"

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	P1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	P2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	P5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	P5-6
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	P6-7 (effectiveness review), P8 (values and preferences) and P8-9 (cost)
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	P7
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix A
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	P7
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	P7
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	P6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	P7
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	P8
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	P8
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	P8
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	P8
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	P8
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	P8

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	P8
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	P8
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	NA
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	P8
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	P9
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	NA
Study characteristics	17	Cite each included study and present its characteristics.	P9-10
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	P12
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	P12
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	P12
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	P10-13
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	P10-13
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	NA
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	P12
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	P18-19
	23b	Discuss any limitations of the evidence included in the review.	P20
	23c	Discuss any limitations of the review processes used.	P20
	23d	Discuss implications of the results for practice, policy, and future research.	P19-20
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	P3,6
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	P3,6
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	P21
Competing	26	Declare any competing interests of review authors.	P21

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
interests			
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	P21

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71
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PRISMA 2020 for Abstracts Checklist

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	Y
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Y
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Y
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Y
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Y
Synthesis of results	6	Specify the methods used to present and synthesise results.	Y
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Y
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Y
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Y
Interpretation	10	Provide a general interpretation of the results and important implications.	Y
OTHER			
Funding	11	Specify the primary source of funding for the review.	Y
Registration	12	Provide the register name and registration number.	Y

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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Self-testing for pregnancy: a systematic review and meta-analysis

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Abstract (292)

Objectives: Urine pregnancy tests are often inaccessible in low-income settings. Expanded provision of home pregnancy testing could support self-care options for sexual and reproductive health and rights. We conducted a systematic review of pregnancy self-testing effectiveness, values and preferences, and cost.

Design: Systematic review and meta-analysis using the GRADE approach.

Data sources: PubMed, CINAHL, LILACS and EMBASE and four trial registries were searched through 2 November 2020.

Eligibility criteria for selecting studies: We included trials and observational studies that compared urine self-testing for pregnancy to health worker-led pregnancy testing on effectiveness outcomes; quantitative and qualitative studies describing values and preferences of end users and health workers; and costs of pregnancy self-testing.

Data extraction and synthesis: Two independent reviewers used standardized methods to search, screen, and code included studies. Risk of bias was assessed using the Cochrane Collaboration and Evidence Project tools. Meta-analysis was conducted using random effects models. Findings were summarized in GRADE evidence profiles and synthesized qualitatively.

Results: For effectiveness, four randomized trials following 5493 individuals after medical abortion showed no difference or improvements in loss to follow-up with home pregnancy self-testing compared to return clinic visits. One additional trial of community health workers (CHWs) offering home pregnancy tests showed a significant increase in pregnancy knowledge and antenatal counseling among 506 clients. Eighteen diverse values and preferences studies found support for pregnancy self-testing because of quick results, convenience, confidentiality/privacy, cost, and accuracy. Most individuals receiving pregnancy self-tests for post-abortion home management preferred this option. No studies reported cost data.

Conclusion: Pregnancy self-testing is acceptable and valued by end users. Effectiveness data comes mostly from articles on post-abortion care, and cost data is lacking. Greater availability of pregnancy self-tests, including in post-abortion care and CHW programs, may lead to improved health outcomes.

Keywords: Pregnancy test, self-care, systematic review, values and preferences, post-abortion care, community health workers, sexual and reproductive health

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Systematic review registration number: PROSPERO CRD42021231656

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Strengths and limitations of this study

- This systematic review and meta-analysis used a comprehensive search for articles not only on the effectiveness of pregnancy self-testing (both randomized trials and comparative observational studies) but also on values and preferences of end users and health workers, and costs.
- We used the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) approach to evaluate the strength and quality of the evidence.
- Because pregnancy self-testing is commonplace in many settings, it is perhaps unsurprising that not many studies have examined the comparative effectiveness of this intervention on home use versus facility-based access, except in specific circumstances like post-abortion care and community health worker programs.
- Values and preferences data came from diverse populations around the world, suggesting high generalizability, but no comparative cost data on pregnancy self-testing were identified.

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Introduction

Urine tests for pregnancy measure the presence of human chorionic gonadotropin (hCG), and are widely used to detect pregnancy in both home and clinical settings. Urine pregnancy tests have evolved substantially since they were first developed 100 years ago.[1] The current generation of tests have close to 100% sensitivity and specificity in detecting hCG at concentrations of 25 milli-IU/ml or more, and thus are able to detect pregnancy as early as one day after a missed menstrual period, if performed per the manufacturers' instructions.[1] However, test performance varies based on characteristics of the test, such as what form of hCG is detected,[2, 3] as well as user characteristics. A systematic review published in 1998 identified five studies that evaluated the diagnostic efficiency of home pregnancy tests.[4] Sensitivity ranged from 0.52 to 1.0 across tests. A gradient of sensitivity was identified by user group. Sensitivity was highest in studies where urine samples obtained by the investigators were tested by volunteers (sensitivity: 0.91, 95% confidence interval [CI], 0.84-0.96) and lower in actual patients who performed the test on their own urine samples (sensitivity 0.75 [95% CI, 0.64-0.85]). Similarly, the test effectiveness score (discriminatory power, where a score of 1.0 or less indicates poor distinctions between pregnant and nonpregnant individuals, and higher scores imply greater effectiveness) was 2.75 (95% CI, 2.3-3.2) for studies where subjects were volunteers but 0.82 (95% CI, 0.4-1.2) for studies with actual patients.

Providing pregnancy tests for home use may have a range of benefits for different populations. In Madagascar, randomized trial data has shown that providing pregnancy tests to community health workers for home distribution can increase both engagement in antenatal care services[5] and contraceptive services, since a negative pregnancy test is necessary before initiating some contraceptive methods.[6] Home pregnancy tests have been shown to be an acceptable and feasible option for follow-up among couples undergoing assisted reproduction.[7] There is also evidence supporting the efficacy, safety, and acceptability of urine pregnancy tests to confirm the effectiveness of a medical abortion instead of an ultrasound.[8, 9] While urine pregnancy self-tests are available over-the-counter in many high- and middle-income settings, in many low-income settings, they may be financially inaccessible to most people outside of public health services, or unavailable altogether, leading individuals with the sole option of health facility-based blood tests to confirm pregnancy. For example, most countries in the Eastern Mediterranean Region have pregnancy self-testing widely available in private pharmacies, particularly in urban settings, and mainly used by the upper socio-economic class due cost and knowledge.[10]

Many people in resource-constrained settings are not able to decide if, how many, and when to have children; increased access to self-care interventions, such as pregnancy self-tests, could support their health decision-making. More widespread efforts to provide pregnancy self-tests that can be used at home or another preferred location could support increased autonomy of individuals as well as support multiple programmatic settings to advance sexual and reproductive health and rights.

We sought to review the literature on home use of pregnancy self-testing as an additional option to facility-based testing. We conducted this systematic review in the context of expanding the evidence base of the 2019 WHO's normative guidance on self-care interventions[11] to include

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new considerations related to sexual and reproductive health. This review was also conducted in the context of the COVID-19 pandemic which continues to result in significant disruptions of essential sexual and reproductive health services. The restrictive measures taken to prevent the spread of COVID-19, particularly lockdowns, have resulted in the need for increased availability and accessibility to this self-care intervention.

Methods

This review addressed the question: should self-testing for pregnancy be available as an additional option to health facility-based testing? We reviewed the extant literature in three areas relevant to answering this question: effectiveness of the intervention, values and preferences of end users and health workers, and cost information. The review followed PRISMA guidelines,[12] and the protocol was published on PROSPERO (CRD42021231656). As this review was conducted to inform a WHO guideline, we followed the WHO Handbook for Guideline Development[13] and used the recommended Grading of Recommendation, Assessment, Development, and Evaluation (GRADE) process[14] to summarize findings. Ethical approval was not required for this systematic review, since all data came from published articles.

Effectiveness review

The effectiveness review was designed according to the population-intervention-comparison-outcomes (PICO) format as follows:

Population: Individuals seeking pregnancy testing
Intervention: Urine self-testing for pregnancy
Comparison: Health worker-led testing for pregnancy (health facility or community clinic with either urine and/or serum test for pregnancy)
Outcomes:
(1) Missed ongoing pregnancy
(2) Appropriate clinical follow-up (counseling, antenatal care visit(s), contraceptive services, abortion services, etc.)
(3) Gestational age at pregnancy awareness (knowledge of pregnancy) and at presentation for antenatal care or abortion
(4) Self-efficacy, self-determination, autonomy, and empowerment
(5) Mental health and well-being (anxiety, stress, self-harm)
(6) Adverse events and social harms (including stigma, discrimination, coercion, violence [including intimate-partner violence, violence from family members or community members, etc.] and breaches of confidentiality), and whether these harms were corrected/had redress available
(7) Device-related issues (e.g. test failure, problems with manufacturing, packaging, labelling, or instructions for use)

All results that were compatible with each outcome domain in each study were extracted.

Inclusion criteria

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To be included in the review, an article had to meet the following criteria:

1) Study design that compared urine self-testing for pregnancy to health worker-led testing for pregnancy (health facility or community services with either urine and/or serum test for pregnancy). This included both randomized controlled trials, non-randomized controlled trials and comparative observational studies (including prospective controlled cohort studies, cross-sectional studies, controlled before-after studies and interrupted time series) that compare individuals who received the intervention to those who did not.
2) Measured one or more of the outcomes listed above.
3) Published in a peer-reviewed journal.

No restrictions were placed based on location of the intervention. No language restrictions were used on the search. Articles in English, French, Spanish, and Chinese were coded directly; articles in other languages were translated.

Search strategy

The following electronic databases were searched through the search date of 2 November 2020: PubMed, CINAHL, LILACS and EMBASE. We searched for ongoing randomized controlled trials (RCTs) through clinicaltrials.gov, the WHO ICTRP, PACTR, and the Australian New Zealand Clinical Trials Registry. Secondary reference searching was also conducted on all studies included in the review. Further, selected experts in the field were contacted to identify additional articles not identified through other search methods. The full search strategy was developed for PubMed and adapted for entry into all computer databases (Appendix A). These search terms were used for the main systematic review (PICO question), for the values and preferences review, and for the cost review.

Screening abstracts

Titles, abstracts, citation information, and descriptor terms of citations identified through the search strategy were screened by a member of the senior study staff. Full text articles were obtained of all selected abstracts, and two independent reviewers assessed all full-text articles for eligibility to determine final study selection. Differences were resolved through consensus.

Data extraction and management

Data were extracted independently by two reviewers using standardized data extraction forms. Differences in data extraction were resolved through consensus, and referral to a senior study team member from WHO when necessary.

The following information was gathered from each included study:

- Study identification: Author(s); type of citation; year of publication
- Study description: Study objectives; location; population characteristics; type of urine pregnancy test; description of self-test access; study design; sample size; follow-up periods and loss to follow-up

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- Outcomes: Analytic approach; outcome measures; comparison groups; effect sizes; confidence intervals; significance levels; conclusions; limitations

For randomized trials, risk of bias was assessed using the Cochrane Collaboration's tool for assessing risk of bias.[15] For non-randomized trials but comparative studies, study rigor was assessed using the Evidence Project 8-item checklist for intervention evaluations.[16]

Data analysis

Data were analyzed according to coding categories and outcomes. Where multiple studies reported the same outcome, meta-analysis was conducted using random-effects models to combine risk ratios with the program Comprehensive Meta-Analysis (CMA).[17] Risk ratios were used directly when provided, and were calculated from the number of events and total sample size in the intervention and comparison groups when not reported. Heterogeneity was calculated using the I-squared statistic.

We planned to stratify all analyses by the following categories/subgroups, where possible:

- Pregnancy self-test point of access (e.g. pharmacy, online, community health worker (CHW) distribution)
- Type of pregnancy test
- Population (e.g. age, marital status, rural/urban)
- Vulnerabilities (e.g. obesity, socio-economic status, poverty, disability, literacy/educational level)
- Gestational age
- High-income versus low or middle-income countries

Data were summarized in GRADE Evidence Profile tables using GRADEPro. Where RCT data were available for a given outcome, we prioritized that over observational data.

Values and preferences review

The same search terms and screening process were used to identify studies to be included in the values and preferences review. Studies were included in this review if they presented primary data examining preferences of individuals regarding urine self-testing for pregnancy. We focused on studies examining the values and preferences of women and adolescent girls who were self-testing for pregnancy or individuals who were potential candidates for such self-testing, but we also included studies examining the values and preferences of health workers and other stakeholders. In this section, we also considered issues related to ability to access (by age, gender, or other factors. These could include legal restrictions around who can access e.g. by age, requirement to be married to purchase, etc. These could also include practical barriers, e.g. is it harder for adolescents to access self-tests because they are not allowed out on their own, they have no money, etc.), informed decision-making, coercion, confidentiality, self-determination, health decision-making to terminate or maintain pregnancy (and discussion with partner if appropriate), and seeking redress. These studies could be qualitative or quantitative in nature, but had to present primary data collection – think pieces and review articles were not

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included. Values and preferences literature was summarized qualitatively and organized by study design and methodology, location, and population.

Cost review

The same search terms and screening process were used to identify studies to be included in the cost review. Studies would have been included in this review if they presented primary data comparing costing, cost-effectiveness, cost-utility, or cost-benefit of the intervention and comparison listed in the PICOs above, or if they presented cost-effectiveness of the intervention as it relates to the PICO outcomes listed above. If cost literature had been found, we would have summarized it qualitatively. We planned to classify cost literature into four categories (health sector costs, other sector costs, patient/family costs, and productivity impacts) and within each category would have organized by study design/methodology, location, and population.

Patient and public involvement

Several of the authors are current or previous users of pregnancy self-tests. Feedback was also received from the WHO patient safety working group. Patients were involved in a global survey of values and preferences conducted to inform the WHO guideline on self-care interventions; they thus play a significant role in the overall recommendation informed by this review.

Results

Our search yielded 414 unique references, of which 62 were retained for full-text review (Figure 1). Ultimately, we identified six that met the inclusion criteria for the effectiveness review, 18 values and preferences studies, and no cost studies.

Figure 1. PRISMA flow chart showing disposition of citations through the search and screening process

Effectiveness review

Overall, six studies met the inclusion criteria for the effectiveness review. This included five RCTs[5, 18-21] and 1 observational study.[22] Table 1 presents descriptive data from the 5 RCTs. Four of the RCTs, conducted in a diverse range of countries (India, Vietnam, Austria, Finland, Norway, Sweden, Moldova, and Uzbekistan) were conducted among 5493 individuals receiving medical abortion.[18-21] These RCTs randomized clients to abortion follow-up with home pregnancy testing and a phone call, versus abortion follow-up with the traditional clinic visit, usually with ultrasound confirmation of successful termination. The fifth RCT, conducted in Madagascar, randomized CHWs to receive pregnancy tests to use with their clients versus the standard of care, which the authors said had historically been pregnancy testing available only at clinics.[5]

Table 1. Description of RCTs included in the effectiveness review

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Study	Location	Sample size	Intervention	Comparator	Outcomes
Women who had a medical abortion					
Iyengar et al., 2015	India (Rajasthan: urban and rural)	N=731	Abortion follow-up with home pregnancy testing and phone call	Abortion follow-up with clinic visit	2. Appropriate clinical follow-up
Ngoc et al., 2014	Vietnam (Ho Chi Minh City and Hanoi)	N=1433			
Oppegaard et al., 2015	Austria (Vienna), Finland (Helsinki), Norway (Oslo), Sweden (Stockholm)	N=929			
Platais et al., 2015	Moldova (Chisinau, Balti, and Drochia), Uzbekistan (Tashkent)	N=2400			
Community health workers and their clients					
Comfort et al., 2019	Madagascar (Aloatra Mangoro, Atsinanana and Analanjirofo)	N=506 CHWs and their clients	Providing CHWs with pregnancy tests to give to clients	Pregnancy testing available at clinics	2. Appropriate clinical follow-up 3. Gestational age at pregnancy awareness (knowledge of pregnancy) and at presentation for antenatal care or abortion

The RCTs provided data for two of our PICO outcomes: 2) Appropriate clinical follow-up, and 3) Gestational age at pregnancy awareness (knowledge of pregnancy) and at presentation for antenatal care or abortion. The non-randomized observational study provided data only for the same outcome under appropriate clinical follow-up, so per our protocol, we present only the RCT data here.

Appropriate clinical follow-up was assessed in the four post-abortion RCTs as loss to follow-up, meaning the client did not return for their follow-up visit or was not able to be contacted by phone. One RCT from Vietnam found dramatically improved follow-up in the pregnancy self-testing arm, with only 0.6% of individuals lost compared with 8.1% lost in the comparison arm.[15] The other three RCTs found no statistically significant differences between study arms.[20, 23, 24] In India, a study of 731 post-abortion participants from urban and rural Rajasthan found loss to follow-up was 3.7% in the self-testing arm compared to 4.8% in the

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clinic arm.[23] In urban Austria, Finland, Norway, and Sweden, loss to follow-up among 929 post-abortion participants was 19.5% in the self-testing arm compared to 22.1% in the clinic arm.[20] Finally, the largest study of 2400 post-abortion participants from Moldova and Uzbekistan found <1% in the self-testing arm compared to no loss to follow-up in the clinic arm.[24]

When the four studies were combined in meta-analysis, there was no significant difference between study arms in loss to follow-up (pooled risk ratio: 0.479, 95% confidence interval 0.155 to 1.480) (Figure 2). Heterogeneity was substantial, with an I-squared of 87. Stratification by high- vs. low- and middle-income settings did not yield meaningful differences; no further planned stratifications were possible, given the small number of similar studies identified for this outcome.

Figure 2. Meta-analysis of loss to follow-up (LTFU) among clients following medical abortion, comparing those receiving follow-up at home versus a return clinic visit

Appropriate clinical follow-up was assessed in the CHW RCT as the mean number of clients at risk of pregnancy who received antenatal counselling at their CHW visit per CHW. This was higher in the intervention group (mean difference: 0.39 clients more, 95% CI: 0.14 more to 0.64 more).

Gestational age at pregnancy awareness (knowledge of pregnancy) and at presentation for antenatal care or abortion was also measured in the CHW RCT as the mean number of clients at risk of pregnancy who knew they were pregnant by the end of the visit per CHW. This was higher in the intervention group (mean difference: 0.86 clients more, 95% CI: 0.59 more to 1.13 more).

Table 2 presents the GRADE evidence profile for the RCT outcomes. The four post-abortion RCTs were rated down for indirectness of the population, as we were interested in all users of pregnancy self-tests while these studies focused on a specific subset of users following abortion. They were also rated down for inconsistency, as meta-analytic results across the four studies showed both the possibility of appreciable benefit and appreciable harm. This resulted in low-certainty evidence. The CHW study was downgraded for risk of bias due to self-reporting of outcomes that may have been affected by self-report.

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Table 2. GRADE evidence profile for effectiveness review studies

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	**Study design**	**Risk of bias**	**Inconsistency**	**Indirectness**	**Imprecision**	**Other considerations**	**self-testing for pregnancy**	**clinic testing**	**Relative (95% CI)**	**Absolute (95% CI)**		
Appropriate clinical follow-up: Lost to follow-up (follow up: mean 2 weeks)												
4	randomised trials	not serious [a]	not serious [b]	serious [c]	serious [d]	none	109/2737 (4.0%)	178/2730 (6.5%)	**RR 0.479** (0.155 to 1.480)	**34 fewer per 1,000** (from 55 fewer to 31 more)	⊕⊕◯◯ LOW	CRITICAL
Appropriate clinical follow-up: Clients at risk of pregnancy who received antenatal counselling at visit (follow up: mean 4 months; assessed with: mean number per CHW)												
1	randomised trials	serious [e]	not serious [f]	not serious	not serious	none	N=260 Mean: 0.67	N=246 Mean: 0.28	MD **0.39 clients more** (0.14 more to 0.64 more)		⊕⊕⊕◯ MODERATE	CRITICAL
Gestational age at pregnancy awareness and at presentation for antenatal care or abortion: Clients at risk of pregnancy who knew they were pregnant by the end of the visit (follow up: mean 4 months; assessed with: mean number per CHW)												
1	randomised trials	serious [e]	not serious [f]	not serious	not serious	none	N=260 Mean: 0.95	N=246 Mean: 0.10	MD **0.86 clients more** (0.59 more to 1.13 more)		⊕⊕⊕◯ MODERATE	CRITICAL

CI: Confidence interval; **RR:** Risk ratio; **MD:** Mean difference; **CHW:** Community health worker

Explanations

a. Risk of bias: Blinding of health workers and participants was not possible due to the nature of the intervention, but outcome was judged to be unaffected by blinding.

b. Inconsistency: Although I-squared of 87 indicates substantial heterogeneity, we did not downgrade for inconsistency because there is likely to be true underlying inconsistency across populations, and in sensitivity analyses, all showed that there was no statistically significant difference between intervention and control groups.

c. Indirectness: Downgraded because population is individuals having medical abortions, while PICO includes both individuals desiring and not desiring pregnancy

d. Imprecision: Downgraded because 95% CI for RR includes both 1 (no effect) AND either appreciable harm (0.75) or appreciable benefit (1.25).

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e. Risk of bias: Downgraded because blinding was not possible due to the nature of the intervention, and the outcome may have been affected by blinding, since it was self-reported on a survey.
f. Inconsistency: This could not be evaluated, as there is only a single study.

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For our outcome of missed ongoing pregnancy, several studies of medical abortion follow-up looked at incomplete abortion. However, we did not consider this outcome measure as "missed" pregnancies, as all were identified through the study protocols, except possibly for those lost to follow-up.

No studies measured our other outcomes of interest: self-efficacy, self-determination, autonomy, and empowerment; mental health and well-being; adverse events and social harms; and device-related issues.

Values and preferences review

Overall, 18 studies were included in the values and preferences review.[7, 18-21, 25-37] Table 3 provides descriptive data and key findings of these studies.

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Table 3. Descriptions and key findings of values and preferences studies

Study and location	Population and sample size (n)	Study design: Methods	Key values and preferences findings
End users – general			
Al-Hassan, 2011 Saudi Arabia (Riyadh)	Pregnancy test kit users attending a family planning clinic, n=120	Quantitative: Cross-sectional survey	Reasons for using the home pregnancy test kits included: quick results (48%), convenience (25%), both speed and convenience (34%), privacy (13%), cost (8%), and accuracy (5%).
Childerhose and MacDonald, 2013 USA (online)	Pregnancy test kit users sharing their stories online, n=99	Qualitative: Stories, web logs, and video logs	Themes included privacy, empowerment, consumer choice, and meaning-making. Home pregnancy testing "temporarily transforms these users into biomedical workers who perform […] knowledge production and medical judgment". Home pregnancy tests can be theorized as "a form of tool use that allows non-experts to produce diagnostic knowledge about their own bodies and health; and as the ongoing biopolitical work that is expected of citizens to produce healthy bodies."
Coons et al., 1990 USA (Lexington, Kentucky)	College students who had used a pregnancy test kit, n=131	Quantitative: Cross-sectional survey	Common reasons for using a home pregnancy test kit included: speed in obtaining results (51.9%), confidentiality (30.5%), convenience (16.8%). Cost (5.3%), accuracy (3.8%), and other (2.3%) were less significant factors.
Pike et al., 2013 UK (Manchester)	Adult female volunteers who were provided different kinds of pregnancy tests, n=111	Quantitative: Cross-sectional survey	After experiencing the use of different test formats, participants preferred midstream-format pregnancy tests. Reasons included hygiene concerns and the need to find a suitable container to collect the sample in when using the cassette or strip tests. Percent of respondents who liked each test (assessed as scoring 1 or 2 on a scale from 1-7 with 1 being most positive) were: strip (8.1%), cassette (3.6%), store-brand midstream visual (34.2%), branded midstream visual (31.5%), branded midstream easy-use visual (77.5%), and branded midstream digital (72.7%).
Robinson, 2020 USA	Adults who had used a pregnancy test kit when in different-sex relationships where they did not want to conceive, n=83	Qualitative: In-depth interviews	For women who do not want to be pregnant, home pregnancy tests have become socially and morally mandatory. The cultural script produces an assumption, impetus, and moral obligation that women must test. Women want to know if there is something inside of them that has the potential to change every aspect of their lives, including their health, their relationships, their living situation, their work, and the lives of their descendants. Women may feel pressure to home-test for pregnancy more than other health conditions because of the time constraints associated with abortion and the unavoidable progression timeline of pregnancy. Women also feel more pressure to test because of the availability of the over-the-counter home pregnancy test. In addition to testing for the condition of pregnancy, the pregnancy test shapes interactions in a variety of areas of their social life including their roles, their relationships, and their responsibilities.
Ross, 2018 Scotland	Partnered adults experiencing a first pregnancy, n=15	Qualitative: Longitudinal in-depth interviews	Participants engaged with pregnancy tests in varying ways, with uses shaping and shaped by their experiences of early pregnancy more broadly. Particular technical characteristics of the home pregnancy test led many participants to question their interpretation of a positive result, as well as the accuracy of the test itself. Rather than addressing the unknowns of early gestation by confirming a suspected pregnancy, a positive result could thus exacerbate uncertainty.
End users – following medical abortion			
Blum et al., 2016 Vietnam (Ho Chi Minh City)	Individuals seeking medical abortion, n=600	Quantitative: Cross-sectional survey at FU	99% of participants found the pregnancy test acceptable in terms of time required to use at home; the rest had no opinion. When asked their preferred location for managing abortion follow-up in the future, 89% said at home with pregnancy test, while 9% said clinic and 2% said no preference.

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Cameron et al., 2012 Scotland (Edinburgh)	Individuals seeking medical abortion, n=75	Quantitative: Cross-sectional survey at FU	All participants said they would be likely to recommend at-home pregnancy testing and phone calls as a follow-up for managing medical abortion to a friend.
Constant et al., 2017 South Africa (Cape Town)	Individuals seeking medical abortion, n=525	Quantitative: Cross-sectional survey at FU	Nearly all participants (98.5%) preferred self-assessment to an in-clinic follow-up appointment. Most were satisfied with their abortion (97.8%), would recommend the method to a friend (82.0%) and would want the same method, should they need another abortion (86.2%).
Dabash et al., 2016 Tunisia (Nabeul, Sousse, Ben Arous, Hamam Lif, and Tunis)	Individuals seeking medical abortion, n=348	Quantitative: Cross-sectional survey at FU	Most participants would consider using the pregnancy test again (97.4%) and would recommend it to a friend (97.7%) for abortion follow-up.
Harper et al., 2002 USA (17 locations)	Individuals seeking medical abortion, n=2121	Quantitative: Cross-sectional survey at FU	Overall, 92% of participants reported that they would be comfortable with home management. In multivariate analysis, those with higher education and white/black vs. Hispanic/Asian individuals were more likely to be comfortable.
Hassoun et al., 2016 France (17 locations nationally)	Individuals seeking medical abortion, n=177	Quantitative: Cross-sectional survey at FU	Regarding acceptability of the urine pregnancy test, 39.9% of participants found it reassuring, 25.8% found it satisfactory, and 4.5% found it both reassuring and satisfactory. Smaller proportions of participants found it unsettling (12.4%), alarming (2.8%). Very small proportions of participants found it both reassuring and unsettling (1.7%), both satisfactory and alarming (0.6%), or both reassuring and alarming (0.6%); 11.8% had no opinion.
Iyengar et al., 2015 India (Rajasthan)	Individuals seeking medical abortion, n=349	Quantitative: Cross-sectional survey at FU	Most participants in the home-pregnancy-assessment group (334/349 [96%]) were satisfied with the method of abortion follow-up.
Iyengar et al., 2016 India (Rajasthan)	Individuals seeking medical abortion, n=20	Qualitative : In-depth interviews	With home-use of misoprostol, women were able to avoid inconvenience of travel, child care, and housework, as well as maintain confidentiality. The use of a low-sensitivity pregnancy test alleviated women's anxieties about retained products; they found it reassuring, and several experienced a sense of relief. They expressed that doing the test was useful, as it saved a visit to the clinic. The majority said they would prefer medical abortion involving a single visit in future.
Ngoc et al., 2014 Vietnam (Ho Chi Minh City and Hanoi)	Individuals seeking medical abortion, n=1328	Quantitative: Cross-sectional survey at FU	In the phone follow-up group, most participants (88.3% [606/686]) indicated that they preferred to complete their follow-up at home with a phone call. In the clinic follow-up group, fewer women (39.9% [256/642]) reported that they would opt for phone follow-up at home in the future.
Oppegaard et al., 2015 Austria (Vienna), Finland (Helsinki), Norway (Oslo), Sweden (Stockholm)	Individuals seeking medical abortion, n=653	Quantitative: Cross-sectional survey at FU	The majority of participants in both the self-assessment and clinic assessment groups said they would prefer self-assessment at home to routine clinical follow-up if they were to have an abortion in future (self: 272 [82%] of 330; clinic: 190 [59%] of 323, p<0·0001).
Platais et al., 2015 Moldova (Chisinau, Balti, and Drochia), Uzbekistan (Tashkent)	Individuals seeking medical abortion, n=2400	Quantitative: Cross-sectional survey at FU	When asked about their preference for future follow-up, most women in the phone group (76.1%) and the plurality of women in the clinic group (48.1%) preferred phone follow-up. 9.6% of the phone group and 29.1% of the clinic group preferred clinic follow-up, and the remainder had no preference.
Shochet et al., 2017 USA (Stanford, CA), Vietnam (Ho Chi Minh City)	Individuals seeking medical abortion, n=87	Quantitative: Cross-sectional survey at FU	Most participants (73.6%) reported being satisfied or very satisfied with using the pregnancy tests at home, 76.7% would want to use home pregnancy tests to monitor for future fertility treatment, and 84.9% would recommend using home pregnancy tests in addition to hospital visits to a friend. 76.7% agreed using home pregnancy tests in lieu of some hospital visits would be more convenient/save time.

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There were 12 quantitative studies (all cross-sectional surveys) and four qualitative studies. For populations, six studies included general pregnancy test users or volunteers, while 12 studies followed individuals after they received a medical abortion with at-home follow-up including a home pregnancy test. No studies were identified with health workers or other stakeholders.

The studies were conducted in diverse locations: USA (n=5), UK (n=3), India (n=2), Vietnam (n=2), and one each in Austria, Finland, France, Moldova, Norway, Saudi Arabia, South Africa, Sweden, Tunisia, and Uzbekistan.

Among general pregnancy test users, most studies found support for pregnancy tests. Reasons why individuals liked pregnancy tests included getting quick results, convenience, confidentiality/privacy, cost, and accuracy. One study found a preference for midstream results tests when individuals were asked to try a range of tests.[29] This same study found that certain types of tests were less preferred due to perceived cleanliness and convenience.

Among individuals having medical abortions, across all studies, most individuals receiving home management with a pregnancy test said they would prefer this option in the future. This ranged from 76.1% (Moldova and Uzbekistan) to 98.5% (South Africa). In two trials with clinic comparison groups, home management was still the most preferred option among participants in the clinic arms. When asked, clear majorities of participants across studies said they found home management acceptable and would recommend it to a friend.

While most values and preferences studies focused on acceptability, ease of use, and reasons for preferring pregnancy self-testing over other options, few studies also presented findings on our other areas of interest: ability to access, informed decision-making, coercion, confidentiality, self-determination, health decision-making, and seeking redress. One study from the USA found themes of privacy, empowerment, consumer choice, and meaning-making among women using pregnancy tests.[22] Other qualitative studies from the USA and Scotland noted that pregnancy tests shape individuals' relationships with their bodies and their social roles, relationships, and responsibilities;[30-31] one theorized that testing is seen as socially and morally mandatory as "the cultural script produces an assumption, impetus, and moral obligation that women must test" to know what is inside their bodies and how it may shape their lives.[30] Finally, one study from India found that managing abortion at home meant that women were able to avoid the inconvenience of travel, childcare, and housework, as well as maintain confidentiality.[28] The pregnancy test alleviated women's anxieties: they found it reassuring, and several experienced a sense of relief.

Cost review

No studies presented primary data examining cost-effectiveness, cost-utility, or cost-benefit for pregnancy self-testing.

Discussion

Pregnancy self-testing has become common practice in many high- and middle-income settings, though is still not widely available in many low-income settings globally. Increased access and availability of pregnancy self-tests for home use can help confirm a pregnancy and engage with the health system for improved reproductive, maternal and newborn health outcomes. Pregnancy

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self-tests are also an important self-care intervention that increase agency and autonomy in sexual and reproductive decision-making. This systematic review is the first to our knowledge to summarize the available literature on effectiveness, values and preferences, and cost for this important intervention.

In the effectiveness review, we found low-certainty evidence from four RCTs following individuals after medical abortion showing no difference in loss to follow-up with self-testing for pregnancy at home compared to a return clinic visit, although individual study results varied, and one did show substantial improvements in loss to follow-up with pregnancy self-testing. Moderate-quality evidence from one RCT showed that when CHWs were given pregnancy tests to provide to clients at home, there was a significant increase in the mean number of clients per CHW at risk of pregnancy who knew they were pregnant by the end of the visit and who received ANC counseling. There was no data on missed ongoing pregnancy; self-efficacy, self-determination, autonomy, and empowerment; mental health and well-being; adverse events and social harms; or device-related issues. Given the ubiquity of self-testing for pregnancy in many settings, it is perhaps unsurprising that this intervention has not been studied widely in a comparative way except in specific circumstances. Given the positive findings in these two settings (post-abortion care and CHW programs), there is evidence that wider use of home pregnancy tests could have beneficial outcomes within health systems. However, fact that almost all data came from post-abortion care settings limits the generalizability of our findings and the conclusions that can be drawn about public health benefits of this intervention. In addition, the certainty of evidence in the context of post-abortion care was limited by the small number of studies combined in meta-analysis and high heterogeneity.

Further, our effectiveness review only compared studies where individuals had access to home pregnancy tests versus facility-based testing. We excluded studies comparing groups of individuals given home pregnancy tests compared to those given nothing, or standard of care. Several trials have examined this latter question, asking whether having a home-pregnancy test on hand would cause women to have a lower threshold of suspicion for pregnancy, to test for pregnancy more often, or to identify pregnancies earlier. For example, one RCT among low-income women in the United States found that participants given home pregnancy tests did suspect pregnancy more often and test for pregnancy more often than control participants.[38] However, another larger RCT in the United States found no difference between arms in mean time at first pregnancy testing or mean gestational weeks at first positive pregnancy test.[39] In both these cases, control participants presumably had at least some access to over-the-counter home pregnancy tests, but results do speak to the potential impact of expanded access to such tests.

For values and preferences literature, 18 studies from diverse populations globally found that most individuals supported pregnancy self-testing. Reasons why individuals liked pregnancy tests included getting quick results, convenience, confidentiality/privacy, cost, and accuracy. Mid-stream urine tests were preferred. Most individuals receiving post-abortion home management with a pregnancy test said they would prefer this option in the future. Although no data from health workers were identified, these positive findings from end users support broader access to pregnancy self-testing.

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Our review identified no comparative cost data on pregnancy self-testing. Clearly, costs of pregnancy self-tests will vary by setting and specific product, but should generally fall within the range of other over-the-counter maternal health products. Considerations of cost should include not only the cost of the test for the end user, but also the full range of health sector costs (e.g. costs due to delayed pregnancy care), other sector costs, and productivity impacts (e.g. labor and workforce issues). Creative ways of expanding access to pregnancy self-testing within existing health care systems, such as using community health workers, would benefit from including cost and cost-effectiveness assessments.

This review had strengths and limitations. While we conducted a thorough and systematic search, we only included peer-reviewed literature; we thus may have missed grey literature. Strengths also included our double extraction of data and inclusion of not only effectiveness literature, but also literature on values and preferences and costs. However, our findings across these three areas are limited by the literature available. We identified no cost data. Importantly, effectiveness data was heavily skewed towards literature on post-abortion care, representing only a fraction of individuals using pregnancy self-tests globally. While our meta-analysis summarizes the findings of the four effectiveness studies, it may provide a false sense of certainty that this represents the true effect. Further, we only had sufficient studies to meta-analyze the outcome “lost to follow-up” after abortion, which provides just one specific way of assessing the overall outcome category of “appropriate clinical follow-up”. Further research on the effectiveness of pregnancy self-testing (especially on outcomes for which we found no comparative data, i.e. missed ongoing pregnancy; self-efficacy, self-determination, autonomy, and empowerment; mental health and well-being; adverse events and social harms; and device-related issues), values and preferences, and costs is needed, particularly in low-resource settings and in contexts where people would benefit from greater accessibility.

Ultimately, the WHO Consolidated guideline on self-care interventions for health and well-being published in 2021 included the following recommendation: "WHO recommends making self-testing for pregnancy available as an additional option to health worker-led testing for pregnancy, for individuals seeking pregnancy testing. (Strong recommendation; very low certainty evidence)".[40]

Conclusion

Pregnancy self-testing is acceptable and valued by end users. Ensuring universal access to pregnancy self-testing may encourage more women and girls to seek early antenatal care, which is a critical opportunity for health workers to deliver care and support during pregnancy, thus contributing to better health outcomes for women, newborn and children. Novel ways of delivering pregnancy self-tests, including through community health worker programs, show promise on improving sexual and reproductive health and rights outcomes, and should be considered for further study and expansion. In the current context of the COVID-19 pandemic, greater availability of pregnancy self-testing can also help maintain the continuity and quality of sexual and reproductive health services.

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Contributions: MN conceptualized the study following discussion with KG and other WHO Regional Advisors from the European and African Regions. CK and PTY designed the protocol, with feedback from MN. PTY ran the database search and oversaw the search, screening, full text review, and data extraction process. CK and PTY drafted the manuscript. All authors reviewed the draft, provided critical review, and read and approved the final manuscript. The corresponding author, as guarantor, accepts full responsibility for the finished article has access to any data and controlled the decision to publish. The corresponding author attests that all listed authors meet the authorship criteria and that no others meeting the criteria have been omitted.

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Competing interests: None declared.

Patient consent for publication: Not required.

Ethics statement: Ethical approval was not required for this systematic review, since all data came from published articles.

Provenance and peer review: Not commissioned; externally peer reviewed.

Data availability statement: All data relevant to the study are included in the article or uploaded as supplementary information. Extracted data are available on request to the corresponding author.

Transparency declaration: The corresponding author, as guarantor, affirms that the manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned (and, if relevant, registered) have been explained.

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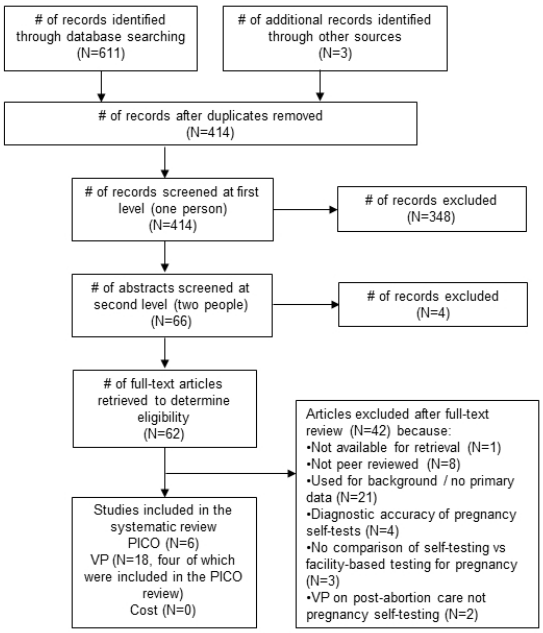


Figure 1. PRISMA flow chart showing disposition of citations through the search and screening process

254x190mm (96 x 96 DPI)

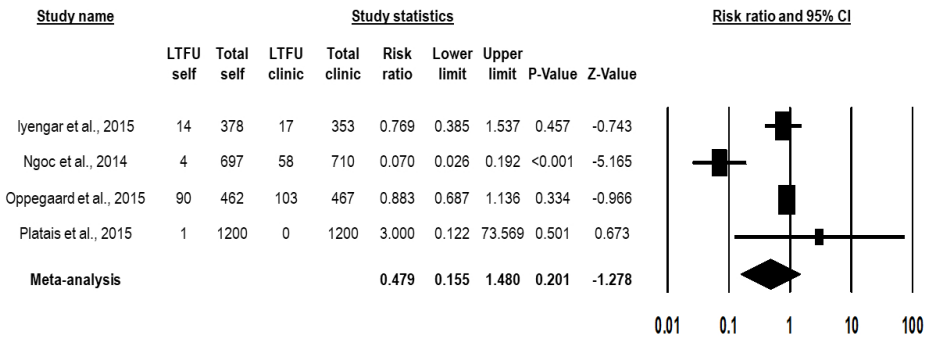


Figure 2. Meta-analysis of loss to follow-up (LTFU) among clients following medical abortion, comparing those receiving follow-up at home versus a return clinic visit

338x190mm (96 x 96 DPI)

Self-testing for pregnancy: a systematic review and meta-analysis
Appendix A. Search Strategy

Search date: 02 Nov 2020

Pubmed
("pregnancy tests" [Mesh] OR "Pregnancy Tests, Immunologic" [Mesh] OR "pregnancy test" [tiab] OR "pregnancy diagnosis" [tiab])

AND

("self care"[Mesh] OR "self-care"[tiab] OR "self-test"[tiab] OR "self-evaluation"[tiab] OR "self-evaluate"[tiab] OR "self-management"[tiab] OR "self-monitor"[tiab] OR "self-manage"[tiab] OR "self-monitored"[tiab] OR "self-managed"[tiab] OR "home"[tiab])

CINAHL
AB: "pregnancy test" OR "pregnancy diagnosis"

AND

AB: "self care" OR "self-test" OR "self-evaluation" OR "self-evaluate" OR "self-management" OR "self-manage" OR "self-monitor" OR "self-monitored" OR "home"

LILACS
"pregnancy test" OR "pregnancy diagnosis"

AND

"self care" OR "self-test" OR "self-evaluation" OR "self-evaluate" OR "self-management" OR "self-manage" OR "self-monitor" OR "self-monitored" OR "home"

EMBASE
('pregnancy test':ti,ab,kw OR 'pregnancy diagnosis':ti,ab,kw) AND ('self care':ti,ab,kw OR 'self-test':ti,ab,kw OR 'self-evaluation':ti,ab,kw OR 'self-evaluate':ti,ab,kw OR 'self-management':ti,ab,kw OR 'self-manage':ti,ab,kw OR 'self-monitor':ti,ab,kw OR 'self-monitored':ti,ab,kw OR 'home':ti,ab,kw)

clinicaltrials.gov
"pregnancy test" OR "pregnancy diagnosis"

WHO ICTRP
"pregnancy test" OR "pregnancy diagnosis"

PACTR
"pregnancy test" OR "pregnancy diagnosis"

Australian New Zealand Clinical Trials Registry
"pregnancy test" OR "pregnancy diagnosis"

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	P1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	P2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	P4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	P4-5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	P5-8 (effectiveness review), P8 (values and preferences) and P8-9 (cost)
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	P6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Appendix A
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	P6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	P6-7
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	P6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	P6-7
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	P7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	P7
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	P7
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	P7
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	P7
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	P7

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	P7
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	P7
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	NA
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	P7
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	P8, Fig 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	NA
Study characteristics	17	Cite each included study and present its characteristics.	P8-9
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	P11
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	P11
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	P11
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	P9-13
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	P9-13
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	NA
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	P12
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	P17-19
	23b	Discuss any limitations of the evidence included in the review.	P19
	23c	Discuss any limitations of the review processes used.	P19
	23d	Discuss implications of the results for practice, policy, and future research.	P19
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	P2,5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	P2,5
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	P20
Competing	26	Declare any competing interests of review authors.	P20

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
interests			
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	P21

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71
For more information, visit: http://www.prisma-statement.org/

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PRISMA 2020 for Abstracts Checklist

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	Y
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Y
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Y
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Y
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Y
Synthesis of results	6	Specify the methods used to present and synthesise results.	Y
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Y
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Y
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Y
Interpretation	10	Provide a general interpretation of the results and important implications.	Y
OTHER			
Funding	11	Specify the primary source of funding for the review.	Y
Registration	12	Provide the register name and registration number.	Y

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